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Reluctant Disclosure and Transparency: Evidence from Environmental Disclosures

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Abstract. Strategic management research increasingly examines firms' strategies for corporate environmental and social disclosures. There are benefits to being perceived as having superior environmental performance, but firms face increasing pressure to provide more complete disclosures, potentially exposing information that will be viewed negatively by external stakeholders. We examine linguistic obfuscation as a means to balance this tension. In particular, we argue that firms may intentionally make their disclosures more complex and harder to understand, thereby blurring the negative content and increasing information processing costs of the recipient. In the context in which an information intermediary actively collects information from firms and evaluates them, we find that firms with unfavorable news to disclose use linguistic obfuscation in information disclosure to manage the tension between the pressure for more complete disclosures and the desire to project a positive image. We further demonstrate that obfuscation lessens the negative impact of reporting negative information on environmental performance ratings given by information intermediaries. This suggests that firms can and do use linguistic tactics to influence environmental ratings.

Keywords: disclosure • obfuscation • ratings • environmental performance • climate change

Introduction

Firms respond strategically to a multitude of institutional pressures. Full compliance with the demands and desires of external stakeholders can bring about enhanced legitimacy and support but may not always be workable or desirable (Oliver 1991). Thus, firms may seek more nuanced responses instead of pursuing full conformity. For example, firms may engage in impression management to respond to institutional pressures and foster a positive image with external stakeholders (Pfeffer 1981, Hiatt et al. 2015). Impression management may be in response to reputational threats (Elsbach 1994, McDonnell and King 2013) or to proactively manage anticipated challenges to organizational practices that are more ambiguously negative (Elsbach et al. 1998). For instance, firms may create strategic noise around a potentially controversial event to distract attention or to prevent causal interpretation between the event and potential negative effects (Graffin et al. 2011).

In this paper, we examine strategic linguistic obfuscation, which we consider as an anticipatory impression management tactic through which firms seek to gain or maintain a positive impression by obfuscating unfavorable news they feel obligated to disclose. Firms are under increasing pressures to be transparent about their business operations. This is especially so in the domain of environmental sustainability, with increasing awareness of climate change and its potentially detrimental effects. Environmental information disclosure, once a niche practice, has become an almost institutionalized practice in recent years. According to the United Nations Global Reporting Initiatives, 92% of the world's largest 250 corporations disclose their sustainability performance.[1] And over 80% of S&P 500 companies published sustainability reports in 2016 (Governance & Accountability Institute 2017).

Increasing environmental disclosure, however, poses challenges for firms. Although environmental disclosures can potentially bring about benefits, such as better reputation and legitimacy (Ullmann 1985, Margolis and Walsh 2003, Cho and Patten 2007), better stakeholder relationships (Tietenberg 1998, Hoffman 2005, Kim and Lyon 2015), and easier access to capital (Flammer 2013, Cheng et al. 2014, Eccles et al. 2014), the benefits of environmental disclosures tend to accrue to those firms that are perceived as superior environmental performers. In fact, firms that reveal poor environmental performance may even be penalized. For example, under the U.S. toxic releases inventory program, firms with poor environmental performance suffered negative stock price shock (Konar and Cohen 1997, 2001), and significant negative stock price returns were measured for firms experiencing environmental crises (Klassen and McLaughlin 1996).

We argue that firms may seek to balance the pressure for more complete environmental disclosures with the desire to protect their reputations by engaging in linguistic obfuscation. Prior work has suggested that under pressure to project a positive image, firms may decouple their external appearance from actual activities (Meyer and Rowan 1977, Zajac and Westphal 2004). For example, firms may communicate positively about poor environmental performance or selectively disclose favorable information (Delmas and Burbano 2011, Kim and Lyon 2015, Marquis et al. 2016). Although some stakeholders, such as regulators, may be susceptible to greenwashing (Kim and Lyon 2015), such strategy can make firms vulnerable to attack from environmental nongovernmental organizations (NGOs) and activists (Kim and Lyon 2015, Marquis et al. 2016). Indeed, a potential backlash imposes such a significant threat that some firms choose not to disclose their environmental achievements to avoid the potential accusation of greenwashing (Carlos and Lewis 2017).

What we study, strategic linguistic obfuscation, is a more subtle linguistic maneuver that involves the use of complex languages and language structures. It is intended to raise the information-processing cost for external stakeholders and make it more difficult for those stakeholders to accurately evaluate the company's environmental performance. This impression management tactic is more delicate than those that involve decoupling. Instead of decoupling, linguistic obfuscation makes it more challenging for external stakeholders to process and interpret the information presented by the firm, and it thus raises the cost of monitoring and scrutiny by stakeholders. Accordingly, it becomes more difficult for external stakeholders to find fault with firms' claims or to accuse them of greenwashing. Certainly, we are not suggesting that linguistic obfuscation is a panacea for companies with unfavorable news to disclose without any risks of backlash. Because linguistic obfuscation raises information processing costs, however, it may delay or reduce any potential negative response by external stakeholders.

We take a two-pronged approach to examine linguistic obfuscation in environmental disclosures. First, we explore drivers of linguistic obfuscation by examining the disclosures made by companies directly to an information intermediary. Second, we examine whether the linguistic obfuscation tactic works by assessing the impact of obfuscation on the environmental performance ratings assigned to the firms by that intermediary. In doing so, we focus on information intermediaries that request and collect environmental information from firms and rate them based on the information disclosed. Because of a lack of consensus on standardized measures of environmental performance, information intermediaries play a critical role in environmental disclosures (and social and governance disclosures) (Luo et al. 2015), filling the institutional void (Dorobantu et al. 2017). In particular, the summary ratings that intermediaries provide serve as a convenient metric informing other external stakeholders because it can be easily understood and compared across firms (Eccles et al. 2011).

We contribute to the extant literature in several respects. First, to the best of our knowledge, the possibility of responding to the increasing demand for more complete disclosures with linguistic obfuscation has not been explored previously. We document that more disclosure does not necessarily mean greater transparency, and we demonstrate a new impression management tactic in the era of increasing corporate information disclosure. This is different from most prior work that studies how firms decide whether to disclose environment information. We investigate how disclosure completeness and disclosure content relate to linguistic obfuscation, conditional on voluntary disclosures. Second, despite the importance of information intermediaries in translating corporate environmental, social, and governance (ESG) performance to easy-to-use summary metrics, not much attention has been paid to ESG intermediaries. Prior research has generally treated ESG ratings given by information intermediaries as objective, beyond the influence of firms' interference. That is, firms disclose their social and environmental performance (Ullmann 1985), then information intermediaries review and rate firms' performance objectively (Sharfman 1996; Chatterji et al. 2009, 2016), and firms respond to ratings (Chatterji and Toffel 2010). However, our research shows that ratings given by information intermediaries can be significantly influenced by firms' managing the linguistic structure of their report. Third, we examine whether and how linguistic obfuscation influences firms' environmental ratings. Prior work has largely left unexplored the impact of impression management tactics on firms' environmental standing. We show that the decrease in environmental performance ratings as a result of negative information revealed is smaller for firms with more obfuscating disclosures. Our findings suggest that a more nuanced form of anticipatory impression management tactic such as linguistic obfuscation, which is hard to identify and criticize, may be effective, at least in the short term.

Strategic Linguistic Obfuscation

Firms actively manage their informational environment, and they do so in ways they hope will favorably affect the impressions of targeted stakeholders (Westphal and Zajac 1998, Graffin et al. 2011). An implicit assumption behind firms' active management

of their information environment is that reactions of observers and markets may be influenced by the actions and tactics of firms (Graffin et al. 2011). In this paper, we examine linguistic obfuscation, by which we mean presenting information in a way that is difficult to understand by virtue of the structure of the linguistic presentation. Firms may deliberately increase the structural complexity of the presented information to preemptively confuse external stakeholders about what conclusions can be drawn from the disclosed information. In this sense, strategic linguistic obfuscation is an anticipatory impression management tactic. As President Harry Truman once said, "If you can't convince them, confuse them."

The strategic use of complex linguistic structures in corporate disclosures is not limited to environmental disclosures. It has long been recognized that corporate documents are hard for outsiders to understand. This may be due to the lack of the technical knowledge to grasp what a well-intentioned firm wishes to convey. Or the firm may deliberately use jargon and complex language structures when the firm does not want outsiders to understand content that it feels compelled to disclose but would rather not reveal. To address the seriousness of this problem in corporate financial disclosures, the U.S. Securities and Exchange Commission (SEC) issued *A Plain English Handbook* in 1998 and has encouraged firms to use "plain English" in disclosure documents. The SEC's criteria for plain English include using short sentences, definitive, concrete, everyday language, the use of the active voice, no legal jargon or highly technical business terms, and no double negatives. Promoting the use of plain English reflects the agency's efforts to make corporate disclosures more meaningful and informative to a wide variety of stakeholders who may or may not necessarily have the expertise to understand complex business languages or jargon. Despite the agency's efforts, evidence shows that strategic linguistic obfuscation continues to plague corporate financial disclosures (Li 2008, Bushee et al. 2018).

Linguistic obfuscation raises the information processing costs of the receiver of the information. Obfuscation is a structure of writing that obscures the information revealed and thereby confuses, distracts, or perplexes readers (Courtis 2004). Through the use of complex verbiage and language structures, obfuscation seeks to mislead by misrepresenting or concealing unfavorable facts. Distorted and confusing writing impedes a reader's comprehension of intended messages. As a result, it is challenging to read and understand such writing and draw conclusive, accurate evaluations. Furthermore, reading complex language structures produces a cognitive load, increasing processing cost and effort (Khawaja et al. 2010). The phonological complexity of written language has been shown in functional MRI studies to increase cognitive activation indicative of higher levels of required attention and cognitive demands in the reader (Desai et al. 2006). In addition, the ease of processing information may affect readers' beliefs about the reliability of information disclosed. A more readable disclosure provides a heuristic cue that information can be relied upon in making related judgments (Shah and Oppenheimer 2007, Häfner and Stapel 2010, Rennekamp 2012).

Readability is especially relevant for environmental reporting, where the problem of information asymmetry between firms and external stakeholders may be more pronounced than in financial reporting because of the lack of consensus on standardized measures of required reports (Luo et al. 2015). With increasing pressure for full disclosures pushing firms to disclose potentially unfavorable news, linguistic obfuscation may present an opportunity to reduce the negative impact of such news on external stakeholders' evaluation of the firm. We explore strategic linguistic obfuscation in corporate environmental reporting under pressure for more complete disclosures.

ExxonMobil's 2016 Corporate Citizenship Report provides an example of what we consider linguistic obfuscation. An excerpt reads as follows:

> In 2016, energy used in our operations totaled 1.5 billion gigajoules. Energy utilized in our operations generates more than 80 percent of our direct greenhouse gas emissions and is one of our largest operating costs. As such, we have focused on energy efficiency for many decades. Since 2000, we have used our *Global Energy Management System* in the Downstream and Chemical businesses, and our *Production Operations Energy Management System* in our Upstream businesses to identify and act on energy savings opportunities. Through our commitment to energy efficiency, application of structured processes and continued use of a bottom-up approach, we continue to yield industry-leading results. (ExxonMobil 2017, p. 20; italics in original)

This statement describes ExxonMobil's energy efficiency efforts, but what ExxonMobil really does is not at all clear to readers, even after multiple readings, because the statement disregards the plain English rule; it uses long sentences and complex words, as well as jargon and highly technical business terms.[2] The appendix provides two further examples, drawn from the text responses to the Carbon Disclosure Project (CDP) survey that we use in the empirical analysis, demonstrating text with relatively high and low levels of linguistic obfuscation. In addition, we had the opportunity to interview two individuals who, in prior positions, were responsible for drafting corporate responses to the CDP survey—one as a consultant specifically hired to improve the client firm's performance

score by improving its survey responses and the other as the employee responsible for environmental disclosures. Both of them described a process through which the responses would be written (and rewritten) and vetted by many members of the organization, with the direct attention of the regulatory and risk compliance team members, who often added conditional statements, caveats, conditions, and explanations to the more sensitive topics in the report. They described a process through which the language in the response became more complex, especially where the company was disclosing unfavorable information, to make the disclosure less damaging. When favorable news was reported, the company's statements were much more straightforward, often shorter, and more direct.

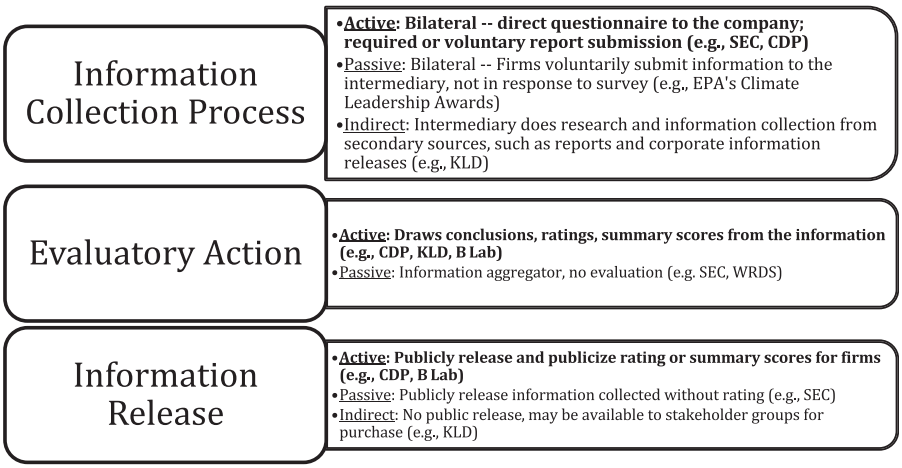
Hypotheses Development

In this paper, we focus on firms' responses to the pressures exerted by information intermediaries that collect and disseminate information on corporate ESG performance. The role of intermediaries as a more objective information source is especially important for ESG performance because greater expertise could be necessary in interpreting more nascent nonfinancial information, relative to standardized financial reporting (Luo et al. 2015). In particular, having easily accessible and usable ESG summary scores provided by intermediaries is of great value to various external stakeholders (Eccles et al. 2011). There are different types of information intermediaries, as described in Figure 1. In this paper, we focus on those that actively collect, evaluate, and disseminate firms' environmental information, reducing the extent of information asymmetry between firms and external stakeholders. Calls for environmental information from active collectors can put significant pressure on firms. Whether to respond to their request is a voluntary decision, yet not doing so may be viewed unfavorably by external stakeholders. Furthermore, especially when there is a bilateral relationship between an active intermediary and the firm from which it is requesting information, there is not much flexibility in reporting because the intermediaries' request is typically well structured, leaving not much room for deciding *what* to disclose. Instead, firms may exercise discretion in *how* to disclose. In particular, we suggest that firms may use linguistic obfuscation to manage the pressure for more complete disclosures and pressures to present a favorable company image through disclosures.

Balancing Pressures to Disclose and Desire to Appear Environmentally Sustainable

Information intermediaries' request to disclose environmental information puts significant pressure on firms. Although environmental disclosures can potentially bring about various benefits, prior studies have shown that the benefits of environmental disclosure tend to accrue to those firms that are perceived as superior environmental performers. For example, there is evidence that securities analysts recommend and investors choose to invest in the stocks of sustainable companies, facilitating access to capital for firms they perceive as more sustainable (Sharfman and Fernando 2008, Dhaliwal et al. 2011, Eccles et al. 2014). Customers may opt for and stay loyal to firms or products that have a more environmentally friendly reputation (Ginsberg and Bloom 2004, Servaes and Tamayo 2013). Local communities may be more cooperative and receptive toward environmentally sound firms (Marquis et al. 2007, Henisz et al. 2014). Employees may be more motivated to work hard for environmentally friendly firms that are good citizens of the society than for those that harm the

Figure 1. Taxonomy of Information Intermediaries



Note. EPA, U.S. Environmental Protection Agency.

environment (Ramus 2001). Nongovernmental organizations and environmental activists may support, or at least not interfere with, the operations of environmentally friendly firms (Sharma and Henriques 2005, Lyon and Maxwell 2011). Government regulators may provide favorable regulatory treatment (Decker 2003, Hiatt and Park 2013, Werner 2015).

This exerts pressure on firms to present and maintain a favorable impression as they disclose more information. Indeed, many firms attempt to appear more environmental friendly than they really are by somehow managing impressions and projecting an environmentally reputable image. For example, this may involve selectively disclosing favorable information (Pfeffer 1981, Kim and Lyon 2011) or communicating positively about their negative performance (Elsbach 1994, Delmas and Burbano 2011). However, firms pursuing this type of impression management tactic face risks of backlash, and recent evidence suggests that firms may understate, or even withhold publicizing, their environmental achievements to avoid charges of greenwashing (Kim and Lyon 2015, Carlos and Lewis 2017). Indeed, some stakeholders, such as activists and NGOs, are keener than other stakeholders on finding out and exposing whether corporate environmental disclosures are mere publicity stunts (Kim and Lyon 2015, Marquis et al. 2016, Carlos and Lewis 2017).

When using the tactic of linguistic obfuscation, targeting information intermediaries that collect environmental information and disseminate summary information to external stakeholders, interactions between firms and these activist stakeholders are more indirect. Thus, this tactic may allow firms to avoid backlash by these skeptical stakeholders while giving the impression of corporate transparency. At the same time, because linguistic obfuscation raises information processing costs and makes it difficult to criticize firms' claims, the likelihood of potential repercussion by information intermediaries may be lower compared with the other impression management tactics described above.

Despite the SEC's efforts to improve disclosure informativeness with *A Plain English Handbook*, academic evidence continues to show that firms strategically manage "readability" in corporate financial disclosures. In particular, poorly performing firms tend to obfuscate the presentation of information, whereas well-performing firms make more straightforward disclosures (Li 2008, Bushee et al. 2018). This general trend has also been found in more specific circumstances. For example, in the context of entrepreneurship, firms that adopt less investor-friendly governance structures use more obfuscation in governance documents for initial public offerings, possibly in an attempt to persuade potential investors that their new ventures are both credible and worthy of funding (Benson et al. 2015). In the context of executive compensation, as the unexplained portion of chief executive officer (CEO) pay increases (relative to a benchmark estimated based on risk-adjusted firm performance), the extent of obfuscation by management in the compensation discussion and analysis section of companies' proxy statements increases, perhaps in an effort to mitigate potential criticisms surrounding compensation (Laksmana et al. 2012).

We posit that the same relationship will hold for environmental disclosures. Responding to the active intermediary's request with information that the firm anticipate will be viewed unfavorably by the intermediary (i.e., reporting "negative news" on one of the key topic areas of importance to the intermediary) would be expected to have a negative impact on the performance ratings earned by the firm. The bilateral relationship between the firm and the information intermediary that characterizes active information collectors means that firms' disclosures can target that intermediary specifically, taking into account what constitutes "negative" or "positive" news for that intermediary. In addition, the fact that the intermediary is both the collector and the evaluator of the information means that firms can employ linguistic tactics in their responses to the information collector as an attempt to influence the rating from this intermediary specifically.

Hypothesis 1. *Firms revealing negative environmental information use more obfuscating language in their disclosure than do firms revealing positive environmental information.*

We now explore the circumstances under which this relationship is more pronounced. We start by exploring the tension between the completeness of disclosure and the desire to maintain a favorable impression. Although much prior work on corporate environmental disclosures has studied whether firms make environmental disclosures, we argue that this binary approach is not as meaningful as it used to be, because stakeholder expectations about corporate environmental sustainability have evolved over time, and most firms make at least perfunctory environmental disclosures of some form these days. Given the changed circumstances, we contend that the extent of disclosure is an important factor to consider in examining corporate environmental sustainability strategy.

When the extent of disclosure is relatively low, meaning firms do not make comprehensive disclosures while disclosing some information, such as through press releases, firms may be able to project a positive image by engaging in framing tactics, attempting to portray company activities in a favorable light. For example, it has been shown that when activists exert pressures through private politics such as

consumer boycotts, protests, and blockades, without resorting to the state or its agencies to effect policy change, firms respond by issuing press releases that frame their actions and motives in ways that protect their image and reputation (Hiatt et al. 2015). Similarly, when firms' reputations are at stake as a result of attacks by activists and NGOs, firms tend to make prosocial claims through press releases in the hope that such claims neutralize or negate the impact of negative attacks (McDonnell and King 2013). The *Oxford English Dictionary* defines a "press release" as an official statement issued to the media giving information on a particular matter. We argue that it is the narrow focus of the disclosure in press releases that makes it possible for firms to positively frame their activities.

Although framing may also work when the extent of disclosure is high, such impression management tactics may have limitations because more complete disclosures provide a more comprehensive picture of company operations related to environmental sustainability. When the firm's activities are more fully exposed, there is less flexibility to present negative information positively. Although firms can still attempt to use framing to positively discuss their actions, more comprehensive disclosure means that they may also need to reveal negative information. This would result in an inconsistent disclosure, with positive framing and negative information, which could invite scrutiny and backlash. Thus, it may not be feasible to thwart potential attacks by engaging in framing tactics. Accordingly, when firms reluctantly make more complete environmental disclosures under pressures to provide more comprehensive information, they may turn to other, more subtle impression management tactics such as linguistic obfuscation.

We contend that obfuscation helps firms manage the tension between more complete disclosures and pressures to create or maintain a positive image. Higher levels of linguistic obfuscation in disclosures increase information processing costs for the external evaluators receiving the information. Obfuscation therefore makes it more difficult for external stakeholders to properly interpret and evaluate company disclosures. Thus, obfuscation allows companies to present the requested information while making it challenging for external stakeholders to clearly identify negative information that can be used to criticize and penalize the company. In this sense, obfuscation is similar to creating strategic noise around a potentially controversial event (Graffin et al. 2011), which also increases information processing costs and limits the ability of external stakeholder to form a clear assessment. But obfuscation is different from strategic noise in that it involves releasing information that is unclear and hard to interpret instead of making multiple releases to blur causal relationships. We posit that when firms are under pressure to provide more complete disclosures and the firms respond by disclosing requested information that reveals negative information, these firms will use more obfuscating language.

Hypothesis 2. *The extent to which firms revealing negative environmental information use obfuscating language in their disclosure increases as they make more complete disclosures.*

Next, we suggest that a firm's prior environmental reputation may influence its incentive to use obfuscation in its disclosure. Reputation plays a critical role in corporate disclosure. Carlos and Lewis (2017) argue that domain-specific reputations, unlike more generalized reputations, may not generate a positive "halo" effect that benefits the company (Fombrun 1996, King 2008). Instead, domain-specific reputations establish expectations in the specific domain, which, if violated, are newsworthy for media outlets to report and more likely to be publicized. Thus, firms with strong domain-specific reputations are more careful in highlighting their high-profile achievements, especially when they face a clear threat that directly contradicts their reputation in the same domain. Such a threat increases the likelihood that external stakeholders detect direct inconsistencies and punish them for perceived hypocritical claims. Because strong domain-specific reputations set high expectations, the negative ramifications of violations can be more severe for firms with strong prior domain-specific reputations than for other firms (Carlos and Lewis 2017).

We contend that even without an imminent domain-specific external threat, such as negative media coverage, firms may expose themselves to reputational threats through information disclosures when the information they are disclosing contradicts a favorable prior reputation. Although much prior work treats firms' environmental performance as a static characteristic, it changes over time (Kim and Lyon 2015). Because corporate environmental performance is dynamic, firms may need to protect their prior reputation when they reveal negative news. Even those firms that once exhibited strong environmental performance may have unfavorable news to report if their environmental performance deteriorates or waivers over time. Various factors can drive this phenomenon. For example, firms may start their environmental sustainability efforts targeting low-hanging-fruit opportunities that offer high financial (and often environmental) paybacks for relatively little initial investment. The firm may garner a positive reputation for taking the initiative to make these investments. However, once such low-cost opportunities are exhausted, corporate environmental efforts may slow down because the marginal cost of achieving further environmental

improvements may be too high to justify additional investments. The failure to engage in further investments shows declining engagement in improving environmental performance. Also, firms may push an aggressive environmental agenda for a while in an effort to differentiate themselves from their competitors. But they may find that their competitive position is eroded instead because of the higher costs incurred to provide environmental differentiation. This may lead to a reversal, abandoning some or all efforts toward environmental sustainability.

Therefore, firms with a favorable reputation for environmental performance may still subsequently report information that indicates less favorable actions. This is especially true as pressure for complete disclosure pushes firms to reveal more information, including negative information that may contradict their good reputation. Thus, we argue that when firms with a strong prior environmental record reveal negative environmental information, putting them at risk for attack, they are more likely to use obfuscation to protect their good reputation in the domain.

Hypothesis 3. *The extent to which firms revealing negative environmental information use obfuscating language in their disclosure as they make more complete disclosures is higher for firms with a good prior environmental reputation.*

Impact of Obfuscation on Ratings

We explore whether strategic linguistic obfuscation succeeds, from the perspective of the firm that is disclosing information. Although prior work demonstrates how firms may use varying forms of impression management tactics in the environmental sustainability domain, to our knowledge, the success or failure of such tactics has rarely been examined. We investigate whether this tactic influences information intermediaries in a way that benefits the firm by securing a more favorable perceived environmental performance. More specifically, the outcome we examine is the environmental performance rating assigned by information intermediaries that actively collect and evaluate information from firms and disseminate a summary performance rating score.

Although limited, there is some evidence on how linguistic obfuscation affects the external stakeholders' perceptions of the company. As mentioned above, for initial public offerings, Benson et al. (2015) show that firms that adopt less investor-friendly governance structures obfuscate more in disclosure documents. The authors further demonstrate that the higher level of obfuscation is associated with raising more capital, suggesting that the negative news regarding governance structures is met with less disfavor when firms are obfuscating more.

We contend that the negative effect of disclosing unfavorable environmental information decreases with higher levels of linguistic obfuscation in the firm's disclosure. Obfuscation operates by making it more challenging and costly for external stakeholders to extract accurate and precise information from firms' disclosures. Negative information, coupled with highly obfuscating linguistic structure, creates uncertainty with regard to what conclusion should be drawn. When the information intermediaries that are the recipients of the information are actively evaluating this information to draw conclusions supporting ratings or scoring, high levels of uncertainty about the true performance of the firm create a gap between the information reported and the rating the firm will receive. By increasing the information processing costs associated with assessing the disclosure and drawing conclusions, linguistic obfuscation makes it difficult for the intermediary to pinpoint negative news and accurately downgrade the company's environmental ratings. Hence, we expect that the penalty for negative news will be smaller for firms employing more linguistic obfuscation.

Hypothesis 4. *The decrease in environmental performance ratings associated with revealing negative information is smaller for firms issuing disclosures with higher levels of linguistic obfuscation.*

We further posit that the impact of linguistic obfuscation on performance ratings differs depending on the firms' prior reputation in the domain of environmental sustainability. Carlos and Lewis (2017) emphasize that firms with strong prior reputations suffer more severe negative consequences from stakeholders in response to apparent inconsistencies between their claims and actions because of the higher expectations that stakeholders have of high reputation firms relative to lower reputation firms. However, we expect that the role that prior reputation plays in information intermediaries forming judgements about firm performance will be different.

The intermediary must figure out how to rate the firm under uncertainty that increases with obfuscation, and the firm's prior reputation in the domain may serve as a positive anchor point (Tversky and Kahneman 1974, Chapman and Johnson 2002). In such a situation, firms with superior domain-specific reputation may get the benefit of the doubt (Fombrun 1996, King 2008) as the intermediary anchors on the (known) prior reputation as a point of reference when uncertainty about current performance is high. Firms with better reputations that submit negative news but use more obfuscating linguistic structures may therefore be penalized less for the negative information. Thus, the degree to which obfuscation buffers firms from the negative impact of disclosing

unfavorable information may be greater for firms with better prior environmental reputations.

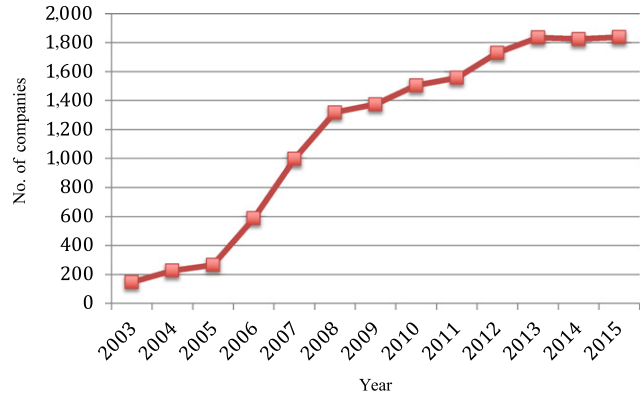
Hypothesis 5. *The extent to which obfuscation lessens the impact of revealing negative information on environmental performance ratings is greater for firms with better prior environmental reputations.*

Data

Our empirical context is the CDP, a nonprofit information intermediary now backed by over 525 large institutional investors that collectively represent $96 trillion in assets as of 2019 (CDP 2019). The CDP is an information intermediary that is an active information collector, analyzing the information collected to create ratings for each firm and publicizing the ratings to other external stakeholders. The purpose is explicitly to provide the investors and other stakeholders with easily accessible and usable information about how climate change will impact the firm and what the firm is doing to mitigate negative impacts and to take advantage of any opportunities. CDP's annual Climate Change Survey of firms began in 2003. Since 2010, the survey questions have been better structured and more fine-grained than in prior years and have been more or less stable. We thus use the CDP data from 2010 to 2015, the last year for which we have access. The CDP surveys all firms that are publicly traded on major exchanges worldwide; Figure 2 depicts the number of publicly traded companies disclosing survey responses to CDP each year.[3] The firms responding to the 2015 survey are from 51 countries and represent 55% of the market capitalization of listed companies globally (CDP 2015a).

We use the data in these voluntary responses to CDP's Climate Change Survey to study disclosure completeness, obfuscation in survey responses, and whether obfuscation affects the performance ratings earned by firms. In the case of the CDP Climate Change Survey, the information intermediary is completely transparent about what responses would constitute negative news in its view. The stated purpose of the CDP, presented on the first line of the survey information request, is "to reduce companies' greenhouse gas emissions (GHG) and mitigate climate change risks," and the CDP publishes a detailed scoring methodology that makes clear that firms will gain points in their evaluation from taking specific actions—most prominently, instituting governance mechanisms to promote response to climate change (more details are provided in the variables section) and making investments to reduce GHG emissions. Our measures of "negative news" being reported by the firm are therefore the absence of governance

Figure 2. (Color online) Number of Publicly Traded Companies Responding to CDP Climate Change Survey by Report Year



mechanisms the firm employs and actions taken by the firm to reduce GHG emissions, drawn directly from the publicly disclosed CDP survey results. We also collect CDP-issued disclosure scores and performance scores for each firm-year. Obviously, we can only include in our analysis the firms that voluntarily disclosed information, so all analyses should be taken as conditional on disclosure. We combine the CDP survey data with firms' financial data and information on stakeholder pressure, detailed below.

Variables. ***Obfuscation: The Fog Index.*** In addition to many standardized questions, the CDP survey asks the respondents to describe how the firm identifies and evaluates climate change-related risks and opportunities that it faces. In particular, subsequent to the yes/no question, "Is climate change integrated into your business strategy?" the company is asked to answer one of the following: "Please describe the process of how climate change is integrated into your business strategy and any outcomes of this process" or "Please explain why climate change is not integrated into your business strategy," depending on its answer to the yes/no question. These follow-up questions are answered in a free text field, and they are the most substantial free text-based answer (i.e., not a standardized answer) in the survey. We use this response to evaluate the level of obfuscation (excerpts from the CDP responses, along with the coding for linguistic obfuscation, are presented in the appendix) .

To measure linguistic obfuscation, we use the "fog index" drawn from the computational linguistics literature. The fog index, developed by Robert Gunning (Gunning 1952), is a well-known metric to measure readability of text. Assuming that the text is well formed and logical, it captures text complexity as a function of syllables per word and words per sentence. The value of the index indicates the number of years

of formal education a reader of average intelligence would need to read the text once and understand the meaning. It is calculated as follows:

$$FogIndex = 0.4\left[\frac{words}{sentence} + 100\left(\frac{complex\ words}{words}\right)\right],$$

where *complex words* are defined as words with three or more syllables. A fog index in the value range of 12–14 is considered ideal.[4] Above that, reading becomes difficult or impossible; below that, it becomes too simplistic or childish.[5]

The fog index is a standard and well-recognized measure for assessing the readability of a document. Former SEC Chairman Christopher Cox discussed the measure in his 2007 speech regarding the abysmal readability of corporate reports to the SEC. He stated that "the Gunning fog index is a venerable linguistic tool that's been around for over half a century. . . . Just as the Black-Scholes model is commonplace when it comes to compliance with the stock option compensation rules, we may soon be looking to the Gunning-Fog and Flesch-Kincaid models to judge the level of compliance with the plain English rules" (Cox 2007).[6] Lehavy et al. (2011), who use the fog index to assess the readability of firm's 10-K filings, state that "the Fog Index has been widely used in social science research for several decades to examine the relation between the readability of written information and various decisions or outcomes" (pp. 1088–1089). They describe the virtues of the index as follows:

> The Fog Index offers several important advantages. First, it allows us to study a large and diverse group of firms. Second, it is an objective measure, not based on analyst surveys or opinion, and can be calculated for any narrative disclosure. Finally, it allows us not only to capture the effect of the readability of a variety of financial items, but, more important, to directly examine the overall syntactic complexity of firms' written public communication, over and above its specific content. (Lehavy et al. 2011, p. 1089)

We use the Lingua::EN:Fathom package of the Perl language to analyze the raw text from the CDP responses.[7] This program has been used in various fields, including information science and business communication (e.g., Collins-Thompson and Callan 2005, Muresan et al. 2006).

Disclosure Completeness. CDP provides an assessment of the completeness of each survey response with the disclosure score, a rating between 0 and 100. The point system assesses the comprehensiveness of the respondent's answers to each of the survey questions[8] and awards a number of points between zero and the maximum possible for that question (the maximum number of points earned for each question varies depending on the importance of the question, as judged by CDP) (CDP 2015b). For text-based (not multiple-choice or numeric answer) responses, the response is judged according to how many topics from a predetermined set are discussed. These topics and the point allocation are described in detail in CDP's scoring methodology, which is available to the respondents (CDP 2015b).

It is important to note that the points for disclosure completeness are assessed based on whether the firm answered the question or provided a specific piece of information, not based on the content of the response. For example, question 2.2 asks whether climate change is integrated into the business strategy of the firm. One point is awarded for answering this multiple-choice question, regardless of whether the answer is "yes" or "no." The follow-up question, answered with free text, asks either how climate change is integrated into the firm's business strategy (if "yes" was selected) or why it is not integrated into the firm's business strategy (if "no" was selected), and the answer for either of these is awarded up to three points, depending on completeness. The total number of points awarded for all questions in the survey is divided by the maximum total points possible, and this ratio multiplied by 100 is the disclosure score. This disclosure score is therefore a good measure of the *completeness* of the survey response.

Performance Score. CDP summarizes the survey responses of each firm with a performance score, a letter grade from A to E, which we code from 5 to 1 (where A equals 5). The score is intended to be a summary of the reported actions to improve the environmental performance of the firm. In particular,

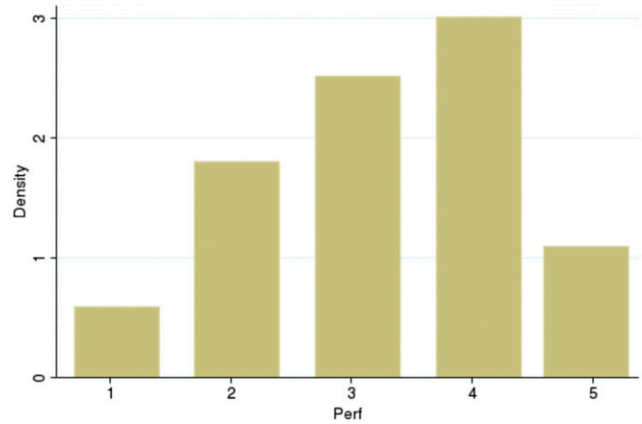
> the performance score assesses the level of action, as reported by the company, on climate change mitigation, adaptation and transparency. Its intent is to highlight positive climate action as demonstrated by a company's CDP response. A high performance score signals that a company is measuring, verifying and managing its carbon footprint, for example, by setting and meeting carbon reduction targets and implementing programs to reduce emissions. (CDP 2015a, p. 12)

CDP's climate change scoring methodology documentation indicates that in calculating the performance score, "Performance points are awarded for evidence provided in the CDP response of actions considered to contribute to climate change mitigation, adaptation and transparency," and "performance points are awarded where a company highlights that it is undertaking, or has undertaken, a 'positive'

climate change action. A positive action is one that contributes to climate change mitigation, adaptation and transparency" (CDP 2015b, p. 3).

It is important to note that the performance score is purposefully designed as an indicator of the degree to which the firm is *acting to improve its environmental impact*, and it is not considered an assessment of the current overall environmental impact of the firm. The scoring methodology states, "It is not a comprehensive metric of the level to which a company is low-carbon or 'green' or sustainable but rather an indication of the level of action taken by the company to manage its impacts on a yearly basis on, and from, climate change" (CDP 2015b, p. 3). In fact, the level of GHG emissions reported in the survey response is not taken into account in the assessment for the performance score (CDP 2015b, p. 35). Performance, from the perspective of the CDP, is thus best understood as efforts to proactively respond to climate change, rather than a rating of the environmental performance (e.g., GHG emissions level) of the firm. The distribution of the performance score is reported graphically in Figure 3.

Figure 3. (Color online) Distribution of Performance Score



Note. A performance score of 5 corresponds to a grade of A, 4 corresponds to a B, and so forth.

Negative Information: Lack of Governance Mechanisms or Actions Taken to Reduce GHG Emissions. Negative news, from the perspective of the CDP, is firms failing to implement the desired actions to respond to and combat climate change. We use responses to several of the multiple-choice questions in the CDP survey to capture the degree to which the firm implements the governance mechanisms and actions that the CDP desires. We code these variables to reflect a failure to implement specific governance mechanisms or actions to create our measures of negative news reported by the firm. On the basis of the question noted above, one measure is *NoProcess*, equal to 1 when the firm reports that it does not integrate climate change into their business strategy. The survey asks whether the firm has instituted an emissions reduction target; we code *NoTarget* equal to 1 when the firm reports not having a target. The survey also asks whether the firm provides managers with incentives to reduce environmental impact, and we code *NoIncentives* equal to 1 for firms that indicate that they do not have such incentives in place. The survey asks which level of the organization is responsible for climate change, with standardized options of board members, executives, lower-level managers, or no one is responsible. We code the latter two options as *No/LowerResponsible* equal to 1, grouping these together because of the small number of responses in these categories. In addition, we explore creating an aggregate metric equal to the count of the governance mechanisms that the firm does have in place: *Governance*, coded as a count from 0 to 4. Finally, the survey asks each firm to report the number of actions taken to reduce greenhouse gas emissions. We code this as *#Actions*, the number of GHG emissions-reducing actions reported.

Reputation. To examine the moderating effect of the firm's domain-specific reputation, we use the prior year performance score rating of the firm. As explained above, we examine the bilateral relationship between the firm and an information intermediary that is an active information collector and evaluator—the CDP. The bilateral relationship allows firms to target the CDP specifically and tailor their response accordingly, especially because the CDP makes clear what constitutes negative or positive news. Thus, in our context, firm reputation vis-à-vis the CDP represents the firm's domain-specific reputation. The prior year performance score rating of the firm is based entirely on the performance criteria dictated by the CDP and thus captures the firm's reputation vis-à-vis the CDP.

Controls. We control for other attributes found in prior work to influence the quality of firm disclosures such as firm size, complex operations, the growth rate of the firm, and the extent of information asymmetry (Li 2008, Dempsey et al. 2012, Laksmana et al. 2012) using annual firm-specific data from the Compustat North America and Global databases, matching this to CDP data using the reported ticker symbol (for the North American data) and ISIN number (for the global data). Specifically, we control for the size of the firm using the number of employees (*#Employees*), availability of capital for investment in environmental performance with the profit margin (i.e., net income divided by sales) (*ProfitMargin*), and the complexity of the firm's operations using the number of countries in which the firm has operations that it includes in its CDP response (*#Countries*). Whereas other studies use the age of the firm to account for the amount of

disclosure that the firm has provided over time, reducing information asymmetry, we use an indicator for whether the firm has responded to the CDP survey in prior years (*PriorYrInCDP*), a more relevant measure of disclosure in this context.[9]

We also experimented with controls for other stakeholder pressure that prior studies have found to influence disclosure strategy. We control for activist pressure using the per-capita count of environmental NGOs (Esty et al. 2005, Marquis et al. 2016), using data from the International Union for Conservation of Nature and country population data from the World Development Indicators from the World Bank. We also control for regulatory pressure with whether the home country has ratified the Kyoto agreement and, if so, whether the home country is one of the Annex 1 countries bound to emissions reductions in the Kyoto agreement. In addition, we include an indicator variable for environmentally sensitive industries, as used by Reid and Toffel (2009) and Cho and Patten (2007); these measures are country and industry specific, respectively. Because our data span countries globally, there are unfortunately no consistent data on firm-specific measures of stakeholder pressure (e.g., shareholder resolutions) that have been used in other studies.

Summary statistics for all of the variables are provided in Table 1. Note that of the governance mechanisms covered in the survey questions, the mechanisms lacking at the largest proportion of responding firms are an emissions target (lacking at 22% of firms) and incentives for managers (lacking at 20% of firms). The average number of actions taken by the respondent firm to improve environmental performance is about 5, although this varies substantially from 0 to 125.

Table 2 reports the pairwise correlations among these variables. From the correlations, it is apparent that firms that fail to implement these governance mechanisms are awarded lower performance scores, and firms that take more actions to improve environmental performance receive higher performance scores, as one would expect. It is important to note that the four variables reflecting governance mechanisms are positively correlated, so that firms implementing one of these mechanisms are likely to also implement others. This is not surprising but has practical implications for interpreting our regression results, which we discuss below. It also motivates the use of the aggregate variable, *Governance*, which reflects a count of how many of these governance mechanisms the firm implements.

Empirical Methodology

We test the two sets of our hypotheses with separate analyses, but all of the tests utilize the pooled cross-sectional data set, with annual observations for each firm respondent. In all cases, we account for the lack of independence across multiple observations for the same firm (in different years) using robust standard errors, clustered at the firm level.

The first analysis tests Hypotheses 1–3. The dependent variable is the natural log of the fog index. The equation to be estimated is as follows:

$$\begin{aligned}\ln(FogIndex_{f,t}) = {} & \beta_1 Governance_{f,t} + \beta_2 \#Actions_{f,t} \\ & + \beta_3 DisclosureScore_{f,t} \\ & + \beta_4 (Governance_{f,t} \times DisclosureScore_{f,t}) \\ & + \beta_5 (\#Actions_{f,t} \times DisclosureScore_{f,t}) \\ & + \beta_6 X_{f,t} + \sum \delta_t + \sum \gamma_i\end{aligned}$$

for firm f in industry i and year t, where δ_t are year indicator variables and γ_i are industry indicator

Table 1. Summary Statistics, CDP Survey Responses: 2010–2015

Variable	*N*	Mean	Std. dev.	Min	Max
FogIndex	5,462	20.75	4.87	0.11	95.28
ln(*FogIndex*)	5,462	3.04	0.34	0.11	4.57
DisclosureScore	5,601	77.46	19.60	0	100
ln(*DisclosureScore*)	5,601	4.31	0.38	0	4.62
PerformanceScore[a]	5,022	3.24	1.11	1	5
NoTarget (0/1)	5,688	0.22	0.42	0	1
No/LowerResponsible (0/1)	6,902	0.03	0.18	0	1
NoIncentives (0/1)	6,902	0.20	0.40	0	1
NoProcess (0/1)	5,688	0.09	0.28	0	1
#Actions	5,462	4.92	5.54	0	125
#Employees ('000s)	6,902	41.32	89.01	0	2,200
ProfitMargin	6,861	0.09	0.39	−16.47	21.71
PriorYrInCDP (0/1)	6,902	0.83	0.38	0	1

Note. Std. dev., Standard deviation.

Table 2. Pairwise Correlations, CDP Survey Responses: 2010–2015

		1	2	3	4	5	6	7	8	9	10
1	*FogIndex*										
2	*DisclosureScore*	0.03									
3	*PerformanceScore*	0.00	0.70								
4	*NoTarget*	0.05	−0.27	−0.41							
5	*No/LowerResponsible*	0.01	−0.11	−0.13	0.11						
6	*NoIncentives*	0.01	−0.35	−0.43	0.29	0.10					
7	*NoProcess*	−0.01	−0.18	−0.26	0.23	0.15	0.26				
8	*#Actions*	0.07	0.12	0.15	−0.05	−0.00	−0.08	−0.07			
9	*#Employees* ('000s)	0.00	0.12	0.17	−0.10	−0.02	−0.11	−0.07	0.08		
10	*ProfitMargin*	0.07	−0.01	−0.01	0.03	−0.01	0.02	0.02	−0.01	−0.08	
11	*PriorYrInCDP*	0.03	0.12	0.13	−0.10	−0.09	−0.08	−0.04	0.04	0.04	0.00

variables, and X is a vector of firm-year level controls. We use ordinary least squares (OLS) to estimate this equation. *Governance* is a vector of the indicators (the lack of each governance mechanisms), or the aggregated count reflecting the four governance mechanisms (i.e., coded as affirmative governance measure), and *#Actions* is the number of emissions-reducing actions taken by the firm during the year reported. Year fixed effects account for common shocks that affect all companies, such as the Intergovernmental Panel on Climate Change issuing a new report detailing the impacts of climate change, the increasing likelihood of widespread carbon pricing regulations, or a significant environmental disaster. Industry fixed effects account for different levels of report complexity, regulation, and disclosure norms across industries and country indicators to account for differences in the language and reporting norms for firms in different geographic locations, as well as country-specific norms related to environmental disclosure and country-level activist pressure (Matisoff et al. 2013).

Hypothesis 1 is supported if β_1 is significantly greater than 0 (or less than 0 for the aggregate governance measure) or β_2 is significantly less than 0, suggesting that firms lacking governance mechanisms and with fewer emissions-reducing actions use more obfuscation in their survey responses. Hypothesis 2, that the relationship between negative news and obfuscation is greater at higher levels of disclosure, is supported if β_4 is greater than 0 (or less than 0 for the aggregate governance measure) or β_5 is less than 0. We test Hypothesis 3, that this relationship is more pronounced for firms with higher reputations, by splitting the sample according to the prior year's performance score for the firm and testing whether the estimated coefficients are significantly larger in magnitude for the high reputation firms.

The second analysis tests Hypotheses 4 and 5. The dependent variable is the encoded performance score, ranging from 1 to 5.[10] The equation to be estimated is as follows:

$$\begin{aligned} PerformanceScore_{f,t} = {} & \beta_1 Governance_{f,t} + \beta_2 \#Actions_{f,t} \\ & + \beta_3 FogIndex_{f,t} \\ & + \beta_4 (Governance_{f,t} \times FogIndex_{f,t}) \\ & + \beta_5 (\#Actions_{f,t} \times FogIndex_{f,t}) \\ & + \beta_6 X_{f,t} + \sum \delta_t + \sum \gamma_i \end{aligned}$$

for firm f in industry i and year t, where δ_t are year indicator variables and γ_i are industry indicator variables, and X is a vector of firm-year level controls. Because of the ordinal nature of the dependent variable, we use an ordered logit estimation and report robust standard errors clustered at the firm level. The *Governance* and *#Actions* variables are the same as above.

The test for Hypothesis 4 is based on the interaction terms. If increased obfuscation decreases the penalty from reporting negative information (as Hypothesis 4 predicts), we would expect β_4 to be greater than 0 and β_5 to be less than 0. We again split the sample by prior reputation and compare the estimated coefficients to test Hypothesis 5. As above, we include year, country, and industry indicators.

Results

The first analysis examines the degree of linguistic obfuscation in the firm's textual response to the CDP survey, testing the predictions (1) that a firm will obfuscate more when it is disclosing negative news (the lack of governance mechanisms and low levels of actions taken); (2) that this will be especially true when the firm is providing a more complete disclosure, as it seeks to manage the tension between more complete disclosure and revealing negative news with increased obfuscation; and (3) that the effect will be larger for firms with preexisting higher reputations. Results are reported in Table 3.

The first column in Table 3 provides the estimates of the fog index as a function of the firm's lack of

Table 3. Obfuscation in CDP Disclosure Increases with Negative News (Dependent Variable: ln(*FogIndex*))

	(1)	(2)	(3)	(4)
NoTarget	0.051**	0.033		
	(0.019)	(0.018)		
No/LowerResponsible	–0.004	–0.014		
	(0.029)	(0.028)		
NoIncentives	0.067**	0.059**		
	(0.020)	(0.019)		
NoProcess	0.145**	0.129**		
	(0.035)	(0.033)		
Governance			–0.056**	–0.054**
			(0.010)	(0.010)
ln(#*Actions*)	–0.018	–0.029**	–0.029**	–0.027**
	(0.009)	(0.009)	(0.009)	(0.010)
EnvSensitive	0.012	0.001	–0.001	
	(0.015)	(0.015)	(0.015)	
PC_NGO	0.046*			
	(0.018)			
Kyoto	–0.004			
	(0.017)			
Annex1	–0.062**			
	(0.017)			
ln(#*Employees*)	–0.013*	–0.011*	–0.010*	–0.014*
	(0.005)	(0.005)	(0.005)	(0.006)
ProfitMargin	0.036	0.000	0.003	–0.005
	(0.028)	(0.027)	(0.027)	(0.028)
ln(*NumCountries*)	0.002	0.004	0.004	–0.004
	(0.005)	(0.005)	(0.005)	(0.005)
ln(*NumWords*)	0.159**	0.151**	0.147**	0.149**
	(0.020)	(0.019)	(0.018)	(0.018)
PriorYrInCDP	–0.006	–0.008	–0.002	0.010
	(0.022)	(0.021)	(0.021)	(0.021)
Fixed effects	Year	Year, country	Year, country	Year, country, industry
R^2	0.28	0.32	0.32	0.35
N	5,252	5,252	5,252	5,252

Notes. Robust standard errors, clustered by firm, are reported in parentheses. The estimation method is OLS. We tested a Heckman selection model correcting for the endogeneity of a firm providing an answer for the text-based survey question used to calculate *FogIndex*. In none of the specifications could we reject the null hypothesis that the equations were independent ($\rho = 0$), meaning that the Heckman selection correction was not justified. Note that all equations that include indicators for target, responsibility, and incentives also include an indicator for surveys that failed to provide a response for each of these questions (less than 1% of the observations for each question).

*5% significance level; **1% significance level.

governance mechanisms in place (individual indicators) and the number of actions taken, with controls for the number of words in the text response;[11] the year of the response; as well as firm-year-level differences in firm size, profitability, the number of countries operated in, whether the firm has reported to the CDP Climate Change survey in a prior year, and country-level controls for the regulatory and institutional pressure to disclosure information. As expected, obfuscation, as measured by the fog index, is higher in the responses of firms with no emissions target in place, no incentives for managers to manage environmental performance, and no process in place to integrate climate change impacts into strategic decision making. The assignment of responsibility for climate change impacts to either no one or a lower-level employee is not a significant predictor of obfuscation, but recall that only 3% of the sample are lacking in this governance mechanism. Firms taking more actions to improve environmental performance obfuscated less in their responses, as expected. These results support our Hypothesis 1, that firms obfuscate more when they are revealing negative news in the survey response. As explained above, we control for regulatory pressure and environmental NGOs at the country level. Regulatory pressure, as reflected by

being located in a country that is part of the Annex 1 list of the Kyoto agreement, is associated with less obfuscation, whereas being in a country with more per-capita environmental NGOs is associated with more obfuscation. As a robustness check, we explore the potential moderating role of these variables in revealing negative news later in this section.

The second column includes country-level fixed effects, controlling not only for differences in regulatory and NGO pressures across countries but also for a host of other differences across countries such as culture and general attitude toward environmental issues. This increases the explanatory power while not materially changing the results of interest. The third column uses the aggregated count of governance measures in place of the four separate (and highly correlated) measures; results are consistent and indicate that the more governance measures the firm is reporting, the lower the level of obfuscation in its report (note that the aggregated measure is coded on the opposite scale of the individual indicators for the *lack* of each governance measure). There is no loss of explanatory power from using the aggregated measure. The final column includes the industry-level fixed effects as well, and it drops the (never significant) indicator for environmentally sensitive industries.

Table 4. Obfuscation Increases with Negative News and Report Completeness (Dependent Variable: ln(*FogIndex*))

	(1)	(2)
Governance	−0.041**	0.206*
	(0.009)	(0.088)
ln(#*Actions*)	−0.014	0.353**
	(0.009)	(0.103)
ln(*DisclosureScore*)	−0.204**	0.018
	(0.047)	(0.059)
Governance × ln(*DisclosureScore*)		−0.058**
		(0.021)
ln(#*Actions*) × ln(*DisclosureScore*)		−0.084**
		(0.024)
ln(#*Employees*)	−0.012*	−0.010
	(0.006)	(0.006)
ProfitMargin	−0.007	−0.013
	(0.026)	(0.029)
ln(*NumCountries*)	−0.002	−0.001
	(0.005)	(0.005)
ln(*NumWords*)	0.167**	0.172**
	(0.020)	(0.020)
PriorYrInCDP	0.027	0.028
	(0.024)	(0.024)
R^2	0.36	0.37
N	4,907	4,907

Notes. Robust standard errors, clustered by firm, are reported in parentheses. All equations contain year, home country, and industry fixed effects.

*5% significance level; **1% significance level.

Results as very similar. This suggests that the level of obfuscation is not primarily driven by the nature of the firm or the country, and it is not limited to firms in specific industries. Estimated coefficients on our explanatory variables of interest change very little with the inclusion of these additional controls.

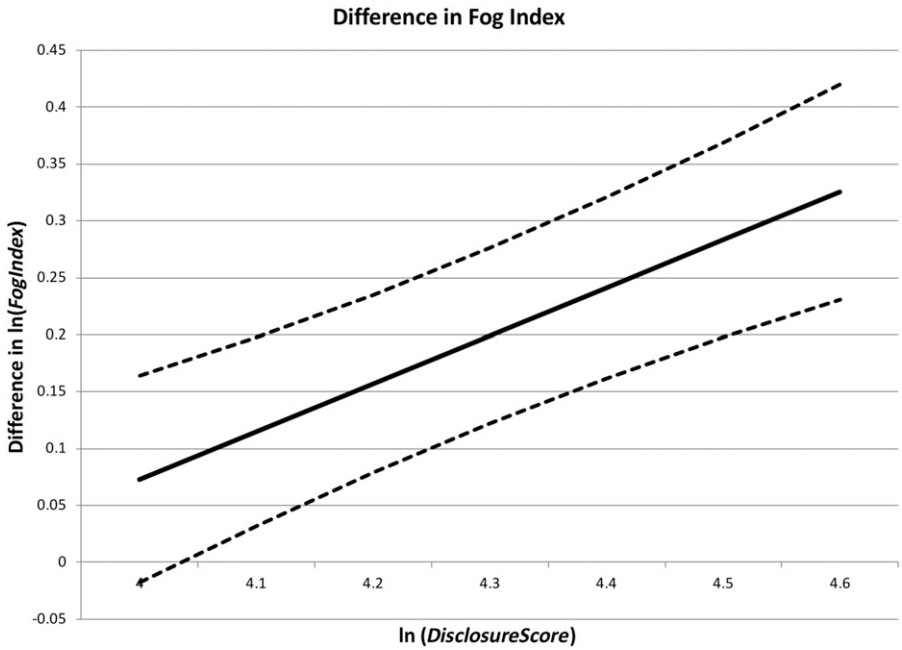
Hypothesis 2 predicts that the relationship between reporting negative news and the levels of obfuscation will be magnified as the firm provides a more complete disclosure. In Table 4, we include the disclosure completeness measure. This variable itself is negatively correlated with obfuscation, and not surprisingly, some of the effect of the governance and actions taken variables is absorbed by the disclosure completeness variable. This is because the reported governance mechanisms and actions taken to improve performance are also substantially correlated with the completeness of disclosure; firms reporting more effort in terms of governance and action taken submit more complete responses.[12]

The test of Hypothesis 2 is reported in column (2), where we include the interactions of disclosure completeness with the governance and the count of actions taken variables. Both of the estimated coefficients are significant and negatively related to the level of obfuscation, indicating that as disclosure completeness increases, firms that reveal negative news submit more obfuscating, less transparent reports.

Figure 4 provides a graphical representations of these results. From the estimated coefficients, we calculated the predicted difference in fog index for firms that reported putting no governance mechanisms in place and taking no actions relative to those firms with all four of the governance mechanisms in place and the average number of actions taken. As the graph shows, the difference in the fog index increases as disclosure completeness goes up. The firms reporting negative news exhibit relatively higher levels of obfuscation in their disclosures at higher levels of disclosure completeness.

The test of Hypothesis 3, that this relationship will be more pronounced for firms with a more favorable prior reputation, is the subsample analysis reported in Table 5. Although the estimated coefficients on the interaction of governance and disclosure completeness appear to be larger and more significant in the high reputation firms (as predicted), the difference from the low reputation firms is not significant. This can be seen most clearly in Figure 5, which graphs the relationship between disclosure completeness and obfuscation for high and low reputation firms reporting negative news (low levels of governance and action). The confidence intervals are overlapping, indicating that the predicted difference between these groups of firms is not supported.

Figure 4. Firms Disclosing Poor Governance and No Actions Taken Provide Reports with Higher Obfuscation as Disclosure Completeness Increases



Notes. Shown is the difference in fog indexes for firms with no governance and no actions *less* firms with all governance attributes and a mean number of actions taken. Dashed lines are 95% confidence interval of estimated difference in the fog index. The mean value of ln(*DisclosureScore*) is 4.36, the median is 4.44, and the 25th percentile is 4.26.

Before moving on to the second set of analyses, we present robustness tests exploring the potential moderating role of regulatory pressure and environmental NGOs in revealing negative news. We include the interactions of these variables with the governance and the number of actions taken in turn. As shown in Table 6, the estimated coefficients for the interaction terms are not significant and do not really impact the estimated results for the disclosure completeness interactions with revealing negative news. As we discuss in more detail in the discussion section, this may be due to more indirect interactions with regulators and activists in our context, whereas the information intermediary is the most salient stakeholder.

The second analysis examines whether this strategy of using obfuscation to manage the tension between the pressure to disclosure more and the revelation of negative information is effective in terms of securing more favorable performance ratings by the CDP. Results for testing Hypothesis 4 are reported in Table 7. As discussed by Hoetker (2007) and Zelner (2009), the interpretation of nonlinear models is complicated by the fact that the predicted effects vary for different levels of the explanatory variables, and the *t*-tests based on estimated standard errors are not reliable indicators of the statistical significance of the estimated coefficients. Accordingly, we follow Hoetker (2007) and Zelner (2009) and provide graphical interpretations of the results, along with confidence intervals, in Figures 6 and 7.

The first column in Table 7 reports the results of the performance score as a function of the indicators for each individual governance mechanism employed by the firm and the number of actions taken to improve environmental performance, controlling for the number of words in the textual response, as well as year of the survey, size, profitability, and home country and industry of the firm. As one would expect, the performance score rating is generally higher for firms that report better governance and for firms that report taking more actions to improve environmental performance. The second column replaces the individual indicators for the governance measure with the aggregate count of governance measures reported (again, recall that this is coded as affirmative governance, whereas the indicators are the *lack* of each governance mechanism). Whereas the governance measure is significant and positive, as expected, the overall explanatory power of the model decreases with this substitution, indicating that the flexibility sacrificed with the aggregate measure limits the model fit. We therefore maintain the individual measure in the subsequent models with interactions while also revisiting the aggregate measure.

The central question of this analysis is whether the reporting of negative news (poor governance or few actions taken to improve) leads to less of a performance

Table 5. Obfuscation as a Function of Negative News Being Reported: By Reputation

	(1) High reputation	(2) High reputation w/ disclosure interaction	(3) Low reputation	(4) Low reputation w/ disclosure interaction
Governance	−0.058**	2.014**	−0.036*	0.306
	(0.021)	(0.776)	(0.015)	(0.184)
ln(#*Actions*)	−0.034**	−0.216	−0.015	0.290
	(0.011)	(0.396)	(0.014)	(0.236)
ln(*DisclosureScore*)		1.152		0.026
		(0.738)		(0.150)
Governance × ln(*DisclosureScore*)		−0.460**		−0.077
		(0.173)		(0.043)
ln(#*Actions*) × ln(*DisclosureScore*)		0.042		−0.069
		(0.087)		(0.053)
ln(#*Employees*)	−0.017*	−0.013	−0.002	−0.001
	(0.008)	(0.008)	(0.010)	(0.010)
ProfitMargin	−0.012	−0.013	0.013	0.006
	(0.050)	(0.052)	(0.048)	(0.049)
ln(*NumCountries*)	−0.002	−0.003	−0.003	0.001
	(0.006)	(0.006)	(0.008)	(0.008)
ln(*NumWords*)	0.235**	0.250**	0.135**	0.154**
	(0.032)	(0.032)	(0.025)	(0.028)
PriorYrInCDP	−0.082	−0.096	0.026	0.028
	(0.062)	(0.063)	(0.038)	(0.038)
R^2	0.50	0.52	0.38	0.40
N	1,753	1,737	1,838	1,796

Note. Robust standard errors, clustered by firm, are in parentheses.
*5% significance level; **1% significance level.

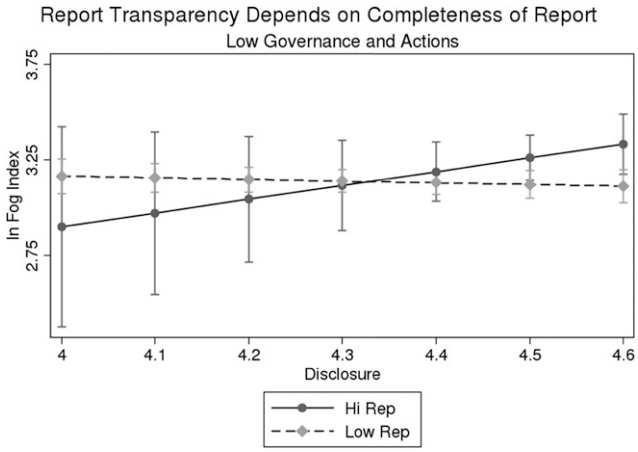
score decrease for the firms that employ higher levels of obfuscation in their report. In other words, does obfuscation soften the blow of negative news, effectively managing the tension between pressures to disclose more information and the negative impact of disclosing negative information? The results in columns (4) and (5) therefore include the interactions of the fog index and the governance measures and number of actions taken to improve.

The estimated coefficients on the interactions indicate that the negative impact of poor governance and few actions taken on the performance score is, in fact, reduced at higher level of obfuscation. Equation (4) utilizes each of the indicators for governance and the interactions of these governance variables and the disclosure score. Figures 6 and 7 are based on these estimates and provide the most rigorous and appropriate means to evaluate the significance and magnitude of the estimated coefficients. The final column reports the interactions using the aggregated governance measure instead of the individual measures. The interaction is negative, as expected, and the coefficient is significant at the 9% level.

The results are presented graphically, based on the "margins" and "marginsplot" commands in Stata, in Figures 6 and 7. Because this is an ordered probit model, each graph represents the probability of a specific outcome. Figure 6 shows the probability of receiving the lowest performance score rating (equal to 1 in our coding) for firms reporting low or high levels of governance, at varying levels of obfuscation. Most important, the likelihood of this low score is reduced for the firms reporting low levels of governance as the degree of obfuscation increases (i.e., the blue line is downward sloping). Firms reporting good governance are less likely to get this bad score, but as their language gets very complex, the likelihood of the low score increases. Figure 7 displays the likelihood of receiving the highest score. Not surprisingly, it is close to zero for firms reporting low governance, but what is interesting is that the likelihood of a high score when reporting low versus high governance levels converges as the level of obfuscation gets very high.

Table 8 reports the results of the final analysis, testing whether this benefit from obfuscation is more pronounced for firms with a favorable preexisting reputation. Again, although the estimated coefficients for the interaction terms of interest appear larger in magnitude and more significant for the high reputation firms, the difference between the two groups is not statistically significant. Figure 8 graphically depicts the likelihood of receiving the lowest performance score, as a function of low governance mechanisms being reported and varying levels of disclosure completeness, for high and low reputation firms. The grey line

Figure 5. Fog Index as a Function of Negative News Being Reported and Report Completeness—Comparing High vs. Low Reputation Firms, Based on Table 5



(high reputation firms) has a steeper slope, but it is not statistically distinguishable from the black line (low reputation firms).

Discussion

There is a growing demand for greater transparency in corporate environmental governance. We have studied the circumstances under which firms respond to this pressure by disclosing more information while engaging in strategic linguistic obfuscation, which increases information processing costs for the recipient of information and makes it difficult to understand the content of disclosure. We have further studied the impact of such obfuscation on the ratings received by the firm.

Our findings demonstrate that firms revealing negative environmental information use more obfuscating language in their disclosure than other firms. Moreover, the extent to which this occurs increases as the firm makes more complete disclosures, consistent with the idea that firms manage the pressure for more disclosure with obfuscation. Despite recent evidence that high reputation firms may face different incentives for impression management around their environmental and social activities, we find no statistically significant difference between high and low prior reputation groups. Regarding the impact of obfuscation, we find that the decrease in environmental performance ratings associated with revealing negative information is smaller for firms with more obfuscating disclosures. In other words, firms benefit from employing linguistic obfuscation when they reveal negative information. The extent to which this occurs is not necessarily greater for firms with good prior environmental performance. That is, a superior prior reputation does not provide a halo effect, although it is also not a great detriment. Below we discuss how these findings extend prior work and prompt new lines of inquiry.

First, our paper builds on and extends an increasing body of work on corporate ESG disclosures—in particular, recent studies that demonstrate more nuanced forms of disclosure strategies. For example, whereas much prior work has focused on how firms with poor environmental performance attempt to project a positive image by communicating positively about their environmental performance, more recent work has started to examine the possibility that firms with good environmental performance may also engage in decoupling behavior, given the multiplicity in the institutional environment. For example, Carlos and Lewis (2017) document that firms that obtain prominent environmental certifications do not always publicize this accomplishment for fear of being perceived as hypocritical and choose to remain silent. Kim and Lyon (2015) argue that when firms do not perform well financially, shareholders may perceive costly actions taken to improve environmental performance as a diversion of resources, and thus, firms may understate their environmental achievements under such circumstances—brownwashing.

We turn our attention back to firms with negative environmental conduct that may engage in more nuanced disclosure tactics, apart from attempting to positively communicate about their poor performance. In doing so, we focus on a situation that is becoming more common: information intermediaries pressuring firms for full disclosure of environmental information. In this case, the firms' ability to elect not to disclose certain information, or to attempt to focus only on favorable information, will be limited. Thus, the firms may seek an alternative means to manage the tension between disclosure and impression management. Responding to pressure for more complete disclosure gives the appearance of transparency, whereas linguistic obfuscation allows firms to blur the negative content disclosed by raising information processing costs and thereby reducing reading comprehension on the part of the recipient. We find that such a tactic works from the perspective of firms when disclosing negative information to information intermediaries. The decrease in environmental performance ratings as a result of negative information revealed is smaller for firms with more obfuscating disclosures. In other words, linguistic obfuscation lowers the extent of negative impact of negative news on environmental ratings.

We further predicted that our findings are likely to be more pronounced for firms with high prior reputations than with low prior reputations. Although our subsample analyses suggested this may be the case, our more in-depth analyses demonstrated that the difference between the two groups is not

Table 6. Robustness Tests Examining Firm Reputation, NGO Intensity, and Regulatory Pressure

	(1)	(2)	(3)	(4)	(5)	(6)
Governance	−0.048**	0.161	−0.043**	0.206*	−0.054**	0.201*
	(0.010)	(0.089)	(0.011)	(0.090)	(0.011)	(0.089)
ln(#*Actions*)	−0.024*	0.321**	−0.019	0.362**	−0.032**	0.358**
	(0.011)	(0.104)	(0.014)	(0.103)	(0.012)	(0.103)
HiReputation	−0.101	−0.128				
	(0.081)	(0.085)				
Governance × *HiReputation*	0.008	0.022				
	(0.020)	(0.021)				
ln(#*Actions*) × *HiReputation*	−0.005	0.001				
	(0.012)	(0.012)				
NGO_Intensity			−2.296**	0.512**		
			(0.226)	(0.153)		
Governance × *NGO_Intensity*			−0.029	−0.025		
			(0.015)	(0.016)		
ln(#*Actions*) × *NGO_Intensity*			−0.024	−0.034		
			(0.019)	(0.019)		
RegulatoryPressure					−0.010	−0.018
					(0.052)	(0.053)
Governance × *RegulatoryPressure*					−0.000	0.007
					(0.012)	(0.013)
ln(#*Actions*) × *RegulatoryPressure*					0.001	−0.004
					(0.015)	(0.015)
ln(*DisclosureScore*)		−0.001		0.008		0.019
		(0.059)		(0.056)		(0.058)
Governance × ln(*DisclosureScore*)		−0.047*		−0.056*		−0.058**
		(0.021)		(0.021)		(0.021)
ln(#*Actions*) × ln(*DisclosureScore*)		−0.077**		−0.084**		−0.085**
		(0.024)		(0.023)		(0.024)
ln(#*Employees*)	−0.010	−0.008	−0.013*	−0.009	−0.014*	−0.010
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
ProfitMargin	0.004	−0.007	−0.006	−0.015	−0.005	−0.014
	(0.028)	(0.029)	(0.027)	(0.029)	(0.028)	(0.029)
ln(*NumCountries*)	−0.003	−0.001	−0.004	−0.002	−0.004	−0.001
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
ln(*NumWords*)	0.155**	0.174**	0.149**	0.172**	0.150**	0.172**
	(0.018)	(0.020)	(0.018)	(0.020)	(0.018)	(0.020)
PriorYrInCDP	0.019	0.033	0.010	0.028	0.010	0.028
	(0.020)	(0.024)	(0.021)	(0.024)	(0.021)	(0.024)
R^2	0.36	0.37	0.35	0.37	0.35	0.37
N	5,252	4,907	5,248	4,906	5,252	4,907

Note. Robust standard errors, clustered by firm, are reported in parentheses.
*5% significance level; **1% significance level.

statistically significant. High reputation firms do not behave fundamentally differently from low reputation firms in making use of linguistic obfuscation. Our results differ from prior work that finds significant effects of domain-specific reputation (Carlos and Lewis 2017) or generalized reputation (McDonnell and King 2013) on corporate ESG disclosure. One explanation is that high reputation firms, anticipating potential backlash from information intermediaries, do not obfuscate more in their reports. Another explanation, as we discuss in more detail below, is that this may be due to differences in the recipient of information. In our paper, the recipient of information is the information intermediary, who presumably works on an objective basis and employs detailed criteria for objective evaluation of firms' disclosures. Thus, for these structured evaluations, prior reputation may not be an important consideration. Indeed, we find no difference in the impact of linguistic obfuscation for firms with high versus low prior reputation in terms of the ratings assigned by the information intermediaries. Our findings suggest that the identity of the recipient of information and the structure of their evaluation processes may be critical factors in

Table 7. Impacts of Obfuscation on Firms' Performance Score: Ordered Probit Estimation (Dependent Variable: *Encoded Performance Score* [1–5, with 5 Equivalent to a Grade of "A"])

	(1)	(2)	(3)	(4)	(5)
NoTarget	–0.998**		–0.978**	–2.225**	
	(0.063)		(0.067)	(0.500)	
No/LowerResponsible	–0.623**		–0.654**	–2.873*	
	(0.143)		(0.142)	(1.171)	
NoIncentives	–0.814**		–0.775**	–1.070*	
	(0.062)		(0.064)	(0.479)	
NoProcess	–0.109		0.150	–1.270	
	(0.098)		(0.105)	(0.990)	
Governance		0.738**			1.264**
		(0.037)			(0.345)
ln(#*Actions*)	0.267**	0.261**	0.217**	–0.213	–0.070
	(0.034)	(0.035)	(0.035)	(0.273)	(0.269)
ln(*FogIndex*)			–0.845**	–1.126**	–0.215
			(0.079)	(0.156)	(0.412)
Interactions with *FogIndex*:					
NoTarget				0.403*	
				(0.160)	
No/LowerResponsible				0.721	
				(0.384)	
NoIncentives				0.095	
				(0.156)	
NoProcess				0.470	
				(0.325)	
ln(#*Actions*)				0.139	0.091
				(0.088)	(0.086)
Governance					–0.191
					(0.111)
ln(#*Employees*)	0.130**	0.132**	0.122**	0.123**	0.126**
	(0.024)	(0.024)	(0.024)	(0.024)	(0.024)
ProfitMargin	0.159	0.152	0.218	0.218	0.204
	(0.136)	(0.138)	(0.138)	(0.137)	(0.140)
ln(*NumWords*)	0.369**	0.348**	0.635**	0.640**	0.587**
	(0.026)	(0.024)	(0.031)	(0.031)	(0.030)
Fixed effects	Year, country, industry	Year, country, industry	Year, country, industry	Year, country, industry	Year, country, industry
Pseudo-R^2	0.23	0.22	0.25	0.25	0.24
N	4,704	4,704	4,576	4,576	4,576

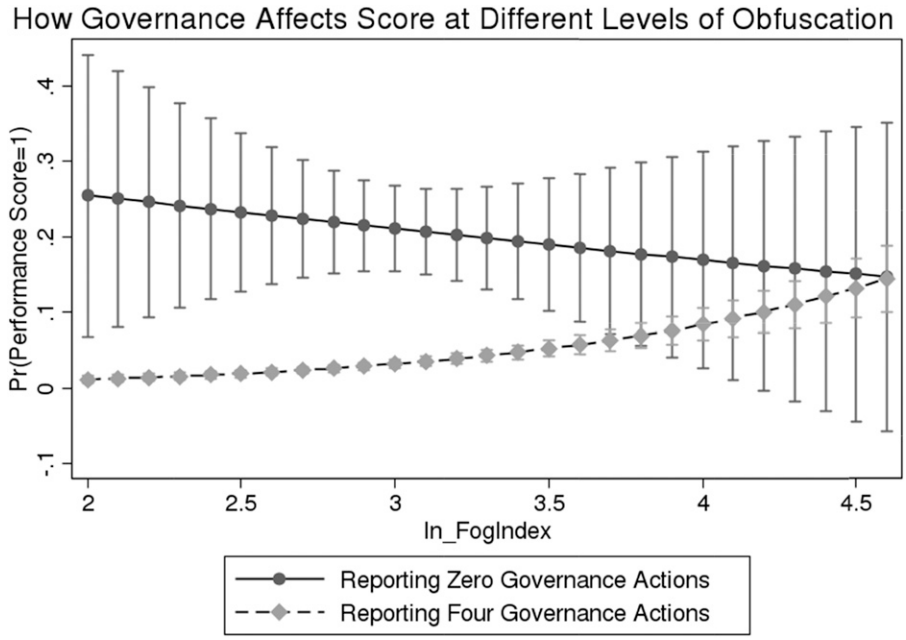
Notes. Robust standard errors, clustered by firm, are reported in parentheses. All equations that include indicators for target, responsibility, and incentives also include indicator for surveys that failed to provide a response for each of these questions (less than 1% of the observations for each question). In Equation (4), a Wald test that the sum of the coefficients on the interactions of governance variables and the disclosure score rejects the null hypothesis that the sum is equal to 0 with a *p*-value of 0.0037. A test for joint significance rejects the null hypothesis that the estimated coefficient on the interactions of governance variables and the disclosure score all equal 0 with a *p*-value of 0.0243.

*5% significance level; **1% significance level.

studying corporate ESG disclosure and exploring the moderating role of corporate reputation.

Our paper also speaks more generally to recent stakeholder research. Dorobantu et al. (2017) argue that stakeholder-oriented efforts can be viewed as ways to create and appropriate value from externalities without internalizing the relevant transactions within the firm, and they propose that such efforts are fundamentally concerned with dealing with weak institutional environments where institutions either are nonexistent or insufficient (incomplete institutions) or are excessive or biased (captured institutions). Our focus on information intermediaries as the recipient of information illustrates how information intermediaries fill institutional voids left under incomplete institutions and how firms attempt to influence the resulting outcome. In this sense, our paper illustrates the importance of third-party institutions in addressing weak institutional environments. Whereas Dorobantu et al. (2017) highlight the role of firms, whether independently or collectively depending on the institutional environment, we argue that discretionary powers of firms may be limited when information intermediaries actively fill the institutional

Figure 6. Likelihood of Performance Score of 1 (Lowest Level)



void. Under the circumstances, firms may engage in more subtle tactics such as linguistic obfuscation that may help them project a positive image or neutralize a negative image through information intermediaries instead of attempting to directly and proactively shape the institutional environment. As described above, there are different types of information intermediaries—some more active than others—collecting and disseminating information. We examined an active information intermediary under incomplete institutions. It would be valuable to study whether and how the nature of information intermediaries influences the findings.

Because of our focus on information intermediaries as the most salient stakeholder, we have considered interactions with other stakeholders only to a limited extent. We explored the possibility that regulators and environmental activists might curtail the impact

Figure 7. Likelihood of Performance Score of 5 (Highest Level)

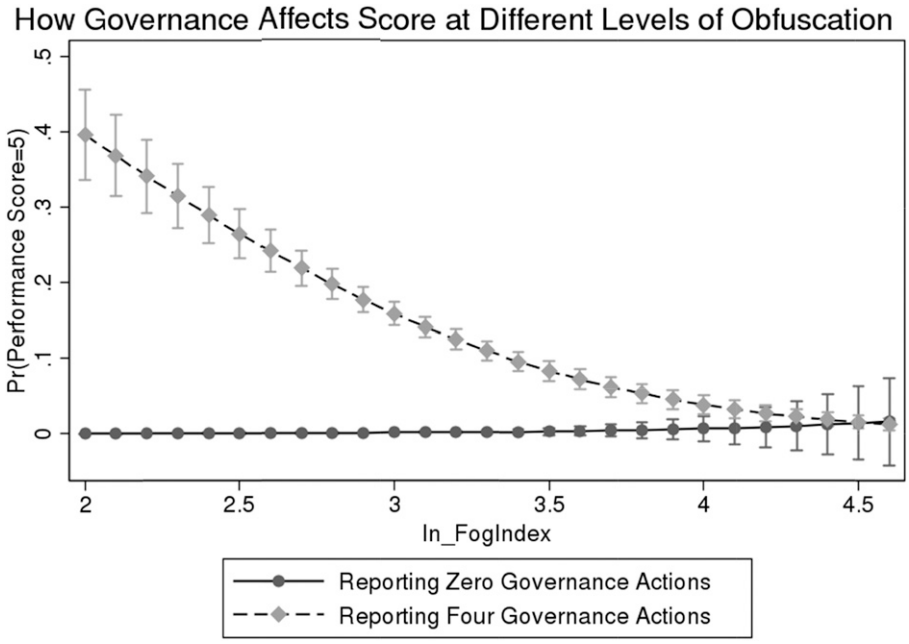


Table 8. Splitting Performance Score Results by Reputation of the Firm

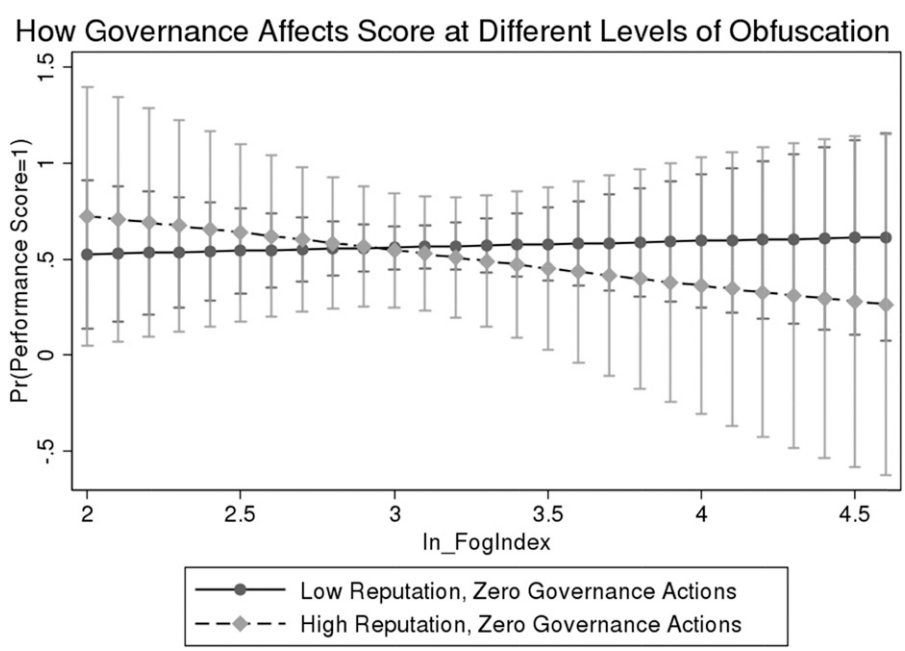
Variable	High reputation	Low reputation
ln(*FogIndex*)	0.619	0.001
	(1.054)	(0.563)
Governance	2.690**	1.103*
	(0.864)	(0.453)
ln(#*Actions*)	–1.155	0.290
	(0.646)	(0.377)
Governance × ln(*FogIndex*)	–0.564*	–0.159
	(0.277)	(0.147)
ln(#*Actions*) × ln(*FogIndex*)	0.426*	–0.030
	(0.209)	(0.121)
ln(#*Employees*)	0.102**	0.052
	(0.036)	(0.032)
ProfitMargin	0.310	0.025
	(0.293)	(0.214)
ln(*NumWords*)	0.588**	0.482**
	(0.059)	(0.037)
N	1,730	1,771

Note. Robust standard errors, clustered by firm, are reported in parentheses.
*5% significance level; **1% significance level.

of disclosing negative information on linguistic obfuscation, and we did not find any significant impacts of these stakeholders. Also, these additional controls did not affect our main findings. Perhaps this is not surprising because, in our setting, interactions between firms and other stakeholders are more indirect, via the summary ratings provided by the information intermediary. Because linguistic obfuscation is not necessarily exposed to these other stakeholders, the possibility of their monitoring and scrutinizing firm behavior may not be as effective at curtailing impression management as when corporate activities are more directly visible to these stakeholders.

Nevertheless, because our exploration of other stakeholders is limited, there may be other stakeholders who can influence the extent to which firms engage in strategic linguistic obfuscation. In particular, we have left unexplored potential influence of owners and top management teams. For example, shareholder resolutions may impose threats on management, and this may in turn influence how firms disclose information (Reid and Toffel 2009, Carlos and Lewis 2017). With shareholders seeking more transparent standardized disclosure regarding climate change, firms may refrain from engaging in linguistic obfuscation considering that disclosures to information intermediaries would ultimately be unveiled to their shareholders. Also, CEO characteristics may influence firm environmental disclosure. For instance, Lewis et al. (2014) find that firms led by CEOs with law degrees are less likely to participate in voluntary environmental disclosures because they tend to obsess about risk and act conservatively. This reasoning suggests that when firms led by CEOs with law degrees face an inevitable situation where they have to disclose negative information, they may engage in more linguistic obfuscation than other firms. In addition, regulators and environmental activists may not be homogeneous stakeholder groups (Lyon 2010, Hiatt et al. 2018), and different types within these stakeholder groups might have a differential

Figure 8. Likelihood of Performance Score of 1 (Lowest Level): Comparison Between High vs. Low Reputation Firms Reporting Negative News as a Function of Fog Index



impact on linguistic obfuscation. Future research along these lines can certainly enhance our understanding of the conditions under which linguistic obfuscation is more prevalent and stakeholder pressures act as deterrents to employing more nuanced disclosure strategy.

Whereas most prior work focuses on environmental disclosure as an impression management tactic, we further examine its impact on environmental ratings. Our finding that linguistic obfuscation lessens the negative impact of disclosing negative information is notable, yet we studied only one type of information intermediaries as the recipient of information. Future research could help refine the boundary conditions of the impression management tactic—linguistic obfuscation—that we explore here.

Acknowledgments

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Appendix. Examples of High and Low Linguistic Obfuscation from the CDP Survey Responses

Low linguistic obfuscation (*FogIndex* = 14.74, number of words = 226, syllables per word = 1.66, number of sentences = 13)	i. The environment and the effects of climate change is identified as one of the core values in Prosafe. As such, key performance indicators have been identified to communicate this. ii. Prosafe has made the strategic decision to implement ISO 14001 within the organisation. Implementation of this standard has resulted in the establishment of goals and objectives to support the strategy. ISO certification was achieved in 2011. iii. Examples of the short term strategy can be demonstrated in the implementation of waste segregation onboard our units. In our offices, we have established targets to reduce the percentage of waste that is disposed of in land fill sites. iv. With regards to Prosafe's long term strategy to reduce its impact on climate change, we have adopted a program for re-engining our existing fleet of vessels. This can be demonstrated by the fact that of the six vessels we operate, two vessels ha[ve] been fitted with new engines in 2009 and 2012, respectively. A third vessel was delivered in 2005 as a new build. v. In Norway, this has provided a strategic advantage, as we reduce the tax burden for our clients when using rigs with low NOX engines. vi. Prosafe has decided on a fleet renewal program. We are building four new build vessels and will ensure that the vessels are as most efficient as practical with regarding to impacts on the environment.
High linguistic obfuscation (*FogIndex* = 28.92, number of words = 441, syllables per word = 2.06, number of sentences = 11)	i. Goals and targets have been established to address climate change. Reporting mechanisms have been put in place to communicate the goals and progress toward the goals throughout the organization, as well as to communicate externally to stakeholders including customers and investors through direct stakeholder requests, our CSR Report, our website and participation in CDP. ii. Climate change aspects that have influenced the strategy include financial implications and impacts on emissions from our global supply chain including our operations, transportation network and ingredient/material sourcing. We have identified those areas in which we can influence or mitigate risks and those areas which require additional monitoring and attention. iii. The most important components of the short term strategy [are] development and implementation of energy conservation, greenhouse gas reduction and education projects, as well as fuel use reduction projects, setting and communicating targets, supply chain communication/engagement, employee education, and communicating our activities, efforts, results and success around climate change to both external and internal stakeholders. iv. The most important components of the long term strategy [are] comprehensive evaluation of financial implications from climate change, development and implementation of alternative energy opportunities, ongoing development and implementation of innovative energy conservation and reduction projects, as well as fuel use reduction projects with an emphasis on alternative energy and fuels, greater emphasis on agricultural impacts, working with suppliers and business partners to understand, monitor and address climate change, and enhanced communication of our activities, efforts, results, successes and engagement opportunities around climate change to both external and internal stakeholders.

Appendix. (Continued)

v. We have made significant strides in reducing greenhouse gas emissions within our operations as a result of reducing the use of electricity and fossil fuels. We recognize these reductions as more than continuous improvement toward increasingly sustainable operations, as we have also benefitted from significant cost savings. We continue to work with the World Cocoa Foundation and the International Center for Tropical Agriculture on improved agricultural practices and evaluation of new regions for cocoa growing to ensure long term security in the availability of quality cocoa for chocolate manufacturing. We are also members of the Sustainability Consortium which provides opportunities for sharing best practices and new technologies to address risk and emissions reductions.

vi. The most substantial business decisions have been around energy conservation and reduction, as well as alternative energy projects which typically achieve higher returns than our average project ROI [return on investment]. We have a flexible payback threshold for energy conservation/reduction projects and evaluate each project on a case-by-case basis to understand the benefits to the business and to the environment. We view this as an investment in our operations while reducing emissions and saving utility costs.

Notes. The text is in response to the following free-text question: "Please describe how your risk and opportunity identification processes are applied at both company and asset level," which is a follow-up question to the multiple-choice question, "Please select the option that best describes your risk management procedures with regard to climate change risks and opportunities" (question CC2.1 in the Strategy-Risk Management Approach section of the questionnaire). Complete text of survey responses can be viewed by searching for respondent company name at https://www.cdp.net/.

Endnotes

[1] See https://www.globalreporting.org/information/sustainability-reporting/pages/gri-standards.aspx. Accessed April 24, 2019.

[2] The fog index of this statement is 21.94 (= 0.4 × [104/4 + 100 × (30/104)]). The fog index is a widely used measure of linguistic complexity, which we use in the paper. Please see the section on variables for more details. A fog index of 21.94 means that the statement requires a graduate degree–level reading comprehension.

[3] For the 2015 report, CDP surveyed 6,316 public companies on major exchanges worldwide and received 2,444 responses, a 39% response rate.

[4] The monotonic transformation scaling the fog index by 0.4 is for the purpose of proving values that roughly correspond to the educational grade level appropriate for the text complexity. For example, a score of 12 is roughly the appropriate level for a 12th grader to read and comprehend.

[5] For this reason, we experimented with including the fog index and its square in all specifications in which it appears, but we did not find that the quadratic specification improved the fit in any case.

[6] We experimented with robustness tests using the Kincaid index in place of the fog index but found that the correlation across these two measures was in excess of 0.98 in our data. The results with this alternative measure were, not surprisingly, nearly identical to those reported here.

[7] See https://metacpan.org/pod/Lingua::EN::Fathom. Accessed April 24, 2019.

[8] The number of survey questions varies each year, but for reference, the 2015 information request was 20 pages long with numerous questions across 15 different sections.

[9] We experimented with controlling instead for the number of years or prior reporting to the CDP, but we found that the indicator for prior reporting was a more significant predictor of disclosure behavior.

[10] This analysis has the added challenge that the CDP assigned a performance score only if the survey response was complete enough; the cutoff set by the CDP was a minimum disclosure score of 50. Therefore, we do not observe the dependent variable for firm-years where the disclosure score is below 50. This is a classic selection problem. We explore addressing this problem using a Heckman selection model, instrumenting for the observation of the performance score using an indicator for whether the firm participated in a mandatory greenhouse gas emissions trading program in the year of the report. The emissions trading programs refer to carbon markets, such as the Emissions Trading Scheme (ETS) in Europe or California's cap-and-trade program in the United States. Firms in these regulatory programs are required to report some environmental performance information—at a minimum, their GHG emissions levels—to a regulatory body on a regular basis (at least annually, but some programs require more frequent reporting). Therefore, firms in these programs are more likely to provide more complete disclosures to the CDP. Yet they do not necessarily have superior environmental performance or ratings because participation in the program is a regulatory mandate, depending only on the firm's location (and whether there is a mandatory ETS program in that location), the industry, and firm size criteria of that specific regulation. One might be concerned that ETS participation, even if exogenously imposed, might be correlated with CDP-assigned performance scores. This is not the case. First, firms that participate in the ETS program do not necessarily reduce GHG emissions. Typically, the program allocates emissions allowances, and thus firms can continue emissions at this level or even purchase emissions allowances in the market for higher levels of emissions. Carbon emission prices have not been high enough to induce changes in firm behavior. Second, the firm's level of GHG emissions is explicitly *not* included in the CDP's evaluation that leads to the performance score (CDP 2015b, p. 35). The CDP survey asks the firms whether they participate in such an ETS program and whether their participation is voluntary or mandatory. We code reported mandatory participation in an ETS program and use this firm-year-level variable as an instrument for the presence of a CDP-assigned performance score, which depends on disclosure completeness meeting the threshold. In the first stage, this instrument is highly predictive of having a CDP-assigned performance score: participating in an ETS increases the likelihood of having a performance score by 52%, and the p-value on the estimated coefficient in less than 0.001. Although this instrument meets the relevance and exclusion requirements, we find that ultimately the test statistics fail to reject the null hypothesis that the equations are independent, indicating that the selection adjustment is unnecessary. The results with the Heckman selection model (available by request from the authors) are nearly identical to those reported here.

[11] As noted above, to separate out the obfuscation from the length of the text response, it is necessary to control for the number of

words in the response when utilizing the fog index as the dependent variable.

[12] This is evident from the correlation statistics (see Table 2) and also in regressions of the disclosure completeness on these variables (results available from authors, not reported in the interest of space).

References

Benson DF, Brau JC, Cicon J, Ferris SP (2015) Strategically camouflaged corporate governance in IPOs: Entrepreneurial masking and impression management. *J. Bus. Venturing* 30(6):839–864.

Bushee BJ, Gow ID, Taylor DJ (2018) Linguistic complexity in firm disclosures: Obfuscation or information? *J. Accounting Res.* 56(1): 85–121.

Carbon Disclosure Project (CDP) (2015a) CDP global climate change report 2015: At the tipping point? Report, CDP, London.

Carbon Disclosure Project (CDP) (2015b) CDP 2015 climate change scoring methodology. Accessed April 24, 2019, https://www.cdp.net/en.

Carbon Disclosure Project (CDP) (2019) About us. Accessed April 24, 2019, https://www.cdp.net/en/info/about-us.

Carlos WC, Lewis BW (2017) Strategic silence: Withholding certification status as a hypocrisy avoidance tactic. *Admin. Sci. Quart.* 63(1):130–169.

Chapman GB, Johnson EJ (2002) Incorporating the irrelevant: Anchors in judgment of belief and value. Gilovich T, Griffin D, Kahneman D, eds. *Heuristics and Biases: The Psychology Of Intuitive Judgment* (Cambridge University Press, Cambridge, UK), 120–138.

Chatterji AK, Toffel MW (2010) How firms respond to being rated. *Strategic Management J.* 31(9):917–945.

Chatterji AK, Levine DI, Toffel MW (2009) How well do social ratings actually measure corporate social responsibility? *J. Econom. Management Strategy* 18(1):125–169.

Chatterji AK, Durand R, Levine DI, Touboul S (2016) Do ratings of firms converge? *Strategic Management J.* 37(8):1597–1614.

Cheng B, Ioannou I, Serafeim G (2014) Corporate social responsibility and access to finance. *Strategic Management J.* 35(1):1–23.

Cho CH, Patten DM (2007) The role of environmental disclosures as tools of legitimacy: A research note. *Accounting Organ. Soc.* 32(7): 639–647.

Collins-Thompson K, Callan J (2005) Predicting reading difficulty with statistical language models. *J. Assoc. Inform. Sci. Techn.* 56(13):1448–1462.

Courtis JK (2004) Corporate report obfuscation: Artefact or phenomenon? *British Accounting Rev.* 36(3):291–312.

Cox C (2007) Closing remarks to the Second Annual Corporate Governance Summit. Speech, Second Annual Corporate Governance Summit, March 23, USC Marshall School of Business, University of Southern California, Los Angeles, http://www.sec.gov/news/speech/2007/spch032307cc.htm.

Decker CS (2003) Corporate environmentalism and environmental statutory permitting. *J. Law Econom.* 46(1):103–129.

Delmas MA, Burbano VC (2011) The drivers of greenwashing. *Calif. Management Rev.* 54(1):64–87.

Dempsey SJ, Harrison DM, Luchtenberg KF, Seiler MJ (2012) Financial opacity and firm performance: The readability of REIT annual reports. *J. Real Estate Finance Econom.* 45(2):450–470.

Desai R, Conant LL, Waldron E, Binder JR (2006) FMRI of past tense processing: The effects of phonological complexity and task difficulty. *J. Cognitive Neuroscience* 18(2):278–297.

Dhaliwal DS, Li OZ, Tsang A, Yang YG (2011) Voluntary nonfinancial disclosure and the cost of equity capital: The initiation of corporate social responsibility reporting. *Accounting Rev.* 86(1):59–100.

Dorobantu S, Kaul A, Zelner B (2017) Nonmarket strategy research through the lens of new institutional economics: An integrative review and future directions. *Strategic Management J.* 38(1): 114–140.

Eccles RG, Ioannou I, Serafeim G (2014) The impact of corporate sustainability on organizational processes and performance. *Management Sci.* 60(11):2835–2857.

Eccles RG, Serafeim G, Krzus MP (2011) Market interest in nonfinancial information. *J. Appl. Corporate Finance* 23(4):113–127.

Elsbach KD (1994) Managing organizational legitimacy in the California cattle industry: The construction and effectiveness of verbal accounts. *Admin. Sci. Quart.* 39(1):57–88.

Elsbach KD, Sutton RI, Principe KE (1998) Averting expected challenges through anticipatory impression management: A study of hospital billing. *Organ. Sci.* 9(1):68–86.

Esty DC, Levy M, Srebotnjak T, Sherbinin A (2005) *2005 Environmental Sustainability Index: Benchmarking National Environmental Stewardship* (Yale Center for Environmental Law and Policy, New Haven, CT), 47–60.

ExxonMobil (2017) 2016 Corporate Citizenship Report. Report, ExxonMobil, Irving, TX, https://corporate.exxonmobil.com/en/~/media/Global/Files/sustainability-report/publication/2016-CCR-full-digital.pdf.

Flammer C (2013) Corporate social responsibility and shareholder reaction: The environmental awareness of investors. *Acad. Management J.* 56(3):758–781.

Fombrun CJ (1996) *Reputation: Realizing Value from the Corporate Image* (Harvard Business School Press, Boston).

Ginsberg JM, Bloom PN (2004) Choosing the right green-marketing strategy. *MIT Sloan Management Rev.* 46(1):79–84.

Governance & Accountability Institute (2017) Flash report: 82% of S&P 500 companies published corporate sustainability reports in 2016. Report, Governance & Accountability Institute, New York.

Graffin SD, Carpenter MA, Boivie S (2011) What's all that (strategic) noise? Anticipatory impression management in CEO succession. *Strategic Management J.* 32(7):748–770.

Gunning R (1952) *The Technique of Clear Writing* (McGraw-Hill, New York).

Häfner M, Stapel D (2010) Information to go: Fluency enhances the usability of primed information. *J. Experiment. Soc. Psych.* 46(1): 73–84.

Henisz WJ, Dorobantu S, Nartey LJ (2014) Spinning gold: The financial returns to stakeholder engagement. *Strategic Management J.* 35(12):1727–1748.

Hiatt SR, Park S (2013) Lords of the harvest: Third-party influence and regulatory approval of genetically modified organisms. *Acad. Management J.* 56(4):923–944.

Hiatt SR, Carlos WC, Sine WD (2018) *Manu militari*: The institutional contingencies of stakeholder relationships on entrepreneurial performance. *Organ. Sci.* 29(4):633–652.

Hiatt SR, Grandy JB, Lee BH (2015) Organizational responses to public and private politics: An analysis of climate change activists and U.S. oil and gas firms. *Organ. Sci.* 26(6):1769–1786.

Hoetker G (2007) The use of logit and probit models in strategic management research: Critical issues. *Strategic Management J.* 28(4):331–343.

Hoffman AJ (2005) Climate change strategy: The business logic behind voluntary greenhouse gas reductions. *Calif. Management Rev.* 47(3):21–46.

Khawaja MA, Chen F, Marcus N (2010) Using language complexity to measure cognitive load for adaptive interaction design. *Proc. 15th Internat. Conf. Intelligent User Interfaces* (ACM, New York), 333–336.

Kim EH, Lyon TP (2011) Strategic environmental disclosure: Evidence from the DOE's voluntary greenhouse gas registry. *J. Environ. Econom. Management* 61(3):311–326.

Kim EH, Lyon TP (2015) Greenwash vs. brownwash: Exaggeration and undue modesty in corporate sustainability disclosure. *Organ. Sci.* 26(3):705–723.

King BG (2008) A political mediation model of corporate response to social movement activism. *Admin. Sci. Quart.* 53(3):395–421.

Klassen RD, McLaughlin CP (1996) The impact of environmental management on firm performance. *Management Sci.* 42(8):1199–1214.

Konar S, Cohen MA (1997) Information as regulation: The effect of community right to know laws on toxic emissions. *J. Environ. Econom. Management* 32(1):109–124.

Konar S, Cohen MA (2001) Does the market value environmental performance? *Rev. Econom. Statist.* 83(2):281–289.

Laksmana I, Tietz W, Yang YW (2012) Compensation discussion and analysis (CD&A): Readability and management obfuscation. *J. Accounting Public Policy* 31(2):185–203.

Lehavy R, Li F, Merkley K (2011) The effect of annual report readability on analyst following and the properties of their earnings forecasts. *Accounting Rev.* 86(3):1087–1115.

Lewis BW, Walls J, Dowell GWS (2014) Differences in degrees: CEO characteristics and firm environmental disclosure. *Strategic Management J.* 35(5):712–722.

Li F (2008) Annual report readability, current earnings, and earnings persistence. *J. Accounting Econom.* 45(2):221–247.

Luo X, Wang H, Raithel S, Zheng Q (2015) Corporate social performance, analyst stock recommendations, and firm future returns. *Strategic Management J.* 36(1):123–136.

Lyon TP (2010) *Good Cop/Bad Cop Environmental NGOs and Their Strategies Toward Business* (Routledge, New York).

Lyon TP, Maxwell JW (2011) Greenwash: Corporate environmental disclosure under threat of audit. *J. Econom. Management Strategy* 20(1):3–41.

Margolis JD, Walsh JP (2003) Misery loves companies: Rethinking social initiatives by business. *Admin. Sci. Quart.* 48(2):268–305.

Marquis C, Glynn MA, Davis GF (2007) Community isomorphism and corporate social action. *Acad. Management Rev.* 32(3):925–945.

Marquis C, Toffel MW, Zhou Y (2016) Scrutiny, norms, and selective disclosure: A global study of greenwashing. *Organ. Sci.* 27(2): 483–504.

Matisoff DC, Noonan DS, O'Brien JJ (2013) Convergence in environmental reporting: Assessing the carbon disclosure project. *Bus. Strategy Environ.* 22(5):285–305.

McDonnell MH, King B (2013) Keeping up appearances: Reputational threat and impression management after social movement boycotts. *Admin. Sci. Quart.* 58(3):387–419.

Meyer JW, Rowan B (1977) Institutionalized organizations: Formal structure as myth and ceremony. *Amer. J. Sociol.* 83(2):340–363.

Muresan G, Cole M, Smith CL, Liu L, Belkin NJ (2006) Does familiarity breed content? Taking account of familiarity with a topic in personalizing information retrieval. *Proc. 39th Hawaii Internat. Conf. System Sci.* (IEEE, Piscataway, NJ), 53c–53c.

Oliver C (1991) Strategic responses to institutional processes. *Acad. Management Rev.* 16(1):145–179.

Pfeffer J (1981) Management as symbolic action: The creation and maintenance of organizational paradigms. Cummings L, Staw B, eds. *Research in Organizational Behavior*, vol. 3 (JAI Press, Greenwich, CT), 1–52.

Ramus CA (2001) Organizational support for employees: Encouraging creative ideas for environmental sustainability. *Calif. Management Rev.* 43(3):85–105.

Reid EM, Toffel MW (2009) Responding to public and private politics: Corporate disclosure of climate change strategies. *Strategic Management J.* 30(11):1157–1178.

Rennekamp K (2012) Processing fluency and investors' reactions to disclosure readability. *J. Accounting Res.* 50(5):1319–1355.

Servaes H, Tamayo A (2013) The impact of corporate social responsibility on firm value: The role of customer awareness. *Management Sci.* 59(5):1045–1061.

Shah A, Oppenheimer D (2007) Easy does it: The role of fluency in cue weighting. *Judgment Decision Making* 2(6):371–379.

Sharfman M (1996) The construct validity of the Kinder, Lydenberg & Domini social performance ratings data. *J. Bus. Ethics* 15(3): 287–296.

Sharfman MP, Fernando CS (2008) Environmental risk management and the cost of capital. *Strategic Management J.* 29(6):569–592.

Sharma S, Henriques I (2005) Stakeholder influences on sustainability practices in the Canadian forest products industry. *Strategic Management J.* 26(2):159–180.

Tietenberg T (1998) Disclosure strategies for pollution control. *Environ. Resource Econom.* 11(3–4):587–602.

Tversky A, Kahneman D (1974) Judgment under uncertainty: Heuristics and biases. *Science* 185(4157):1124–1131.

Ullmann AA (1985) Data in search of a theory: A critical examination of the relationships among social performance, social disclosure, and economic performance of U.S. firms. *Acad. Management Rev.* 10(3):540–557.

Werner T (2015) Gaining access by doing good: The effect of sociopolitical reputation on firm participation in public policy making. *Management Sci.* 61(8):1989–2011.

Westphal JD, Zajac EJ (1998) The symbolic management of stockholders: Corporate governance reforms and shareholder reactions. *Admin. Sci. Quart.* 43(1):127–153.

Zajac EJ, Westphal JD (2004) The social construction of market value: Institutionalization and learning perspectives on stock market reactions. *Amer. Sociol. Rev.* 69(3):433–457.

Zelner BA (2009) Using simulations to interpret results from logit, probit, and other nonlinear models. *Strategic Management J.* 30(12): 1335–1348.

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